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**Explainable AI for Predicting and Analyzing
Cardiovascular Disease Risk**

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Explainable AI for Predicting and Analyzing Cardiovascular Disease Risk

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Abbreviations

CVD(s)	<i>Cardiovascular Disease(s)</i>
CHD	<i>Coronary Heart Disease</i>
ML	<i>Machine Learning</i>
AI	<i>Artificial Intelligence</i>
XAI	<i>Explainable Artificial Intelligence</i>
ASCVD	<i>Atherosclerotic Cardiovascular Disease</i>
FRS	<i>Framingham Risk Score</i>
PCEs	<i>Pooled Cohort Equations</i>
SHAP	<i>SHapley Additive exPlanations</i>
LIME	<i>Local Interpretable Model-agnostic Explanations</i>
PDP	<i>Partial Dependence Plot</i>
ICE	<i>Individual Conditional Expectation (curve)</i>
ALE	<i>Accumulated Local Effects (plot)</i>
LR	<i>Logistic Regression</i>
RF	<i>Random Forest</i>
SVM	<i>Support Vector Machine</i>
kNN	<i>k-Nearest Neighbors</i>
XGBoost/XGB	<i>eXtreme Gradient Boosting</i>
CatBoost/CB	<i>Categorical Boosting</i>
AUC	<i>Area Under the Curve</i>
ROC	<i>Receiver Operating Characteristic (curve)</i>
GB	<i>Gradient Boosting</i>
DL	<i>Deep Learning</i>
Nested CV	<i>Nested Cross-Validation</i>

Abstract

Cardiovascular disease (CVD) remains one of the most difficult problem in global healthcare. Diseases in this category have a direct impact on the patient's mortality. Thus, there is currently a need for early and accurate detection of cardiac disease. Yet, many high-performance machine learning (ML) models are hindered by their "black-box" nature, which limits clinician trust and adoption in clinical practice. This study aims to examine how explainable artificial intelligence (XAI) methods – specifically SHAP – provide interpretability in CVD prediction and analysis. Based on two available datasets – the Cleveland Heart Disease and the Framingham dataset, consisting of CVD medical records, six ML algorithms – Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), Random Forest (RF), eXtreme Gradient Boosting (XGBoost/XGB), and Categorical Boosting (CatBoost/CB), were utilized and evaluated. CatBoost was chosen as best model to apply XAI for explanations after a process of analysis and evaluation with nested CV, calibration and threshold tuning, with the ROC-AUC in the final evaluation of Cleveland dataset reached 0.95 and PR-AUC of 0.27. Interpretability was provided through SHAP global and local plots. SHAP plots showed that, for prognosis, three most important factors are age, gender and blood pressure. On the contrary, diagnosis considers results of noninvasive tests – exercise thallium scintigraphy and fluoroscopy for coronary calcium (thal and ca), and chest pain type (cp), as most influence factors to the model's predictions. SHAP plots provide a clear visualization in how different the models think for different medical needs. ALE and Bootstrapping methods were applied to further validate explanations reliability, in terms of consistency and stability.

Chapter 1 . INTRODUCTION

1.1 Background and Motivation

Cardiovascular diseases (CVDs) remain the formidable leading cause of morbidity and mortality globally. According to the World Health Organization (WHO, 2025), CVDs are responsible for an estimated 19.8 million deaths annually, representing 32% of all global deaths. The burden of these diseases is not only measured in mortality rates, but also in the economic pressure they place on healthcare systems worldwide (Shih, et al., 2025; Kazi, et al., 2024). As a result, early identification of individuals at risk is an important objective in preventive cardiology (WHO, 2025).

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have significantly improved predictive modeling in healthcare (Javaid, Haleem, Singh, Suman, & Rab, 2022). Compared to traditional risk prediction models (discussed in Chapter 2), advanced ML algorithms can capture complex nonlinear relationships among clinical variables and often achieve higher predictive performance, however, many high-performing models operate as “black boxes” (Sanchula, 2025), making it difficult for clinicians to understand how predictions are generated.

In healthcare, just considering predictive accuracy is insufficient. Clinicians must also be able to interpret and justify model outputs before incorporating them into clinical decision-making. Explainable Artificial Intelligence (XAI) has emerged as an approach for improving transparency and trust in ML systems.

This study investigates how XAI can be used to explain CVD prediction in two distinct clinical scenarios: diagnosis and prognosis. By analyzing the Cleveland Heart Disease and Framingham datasets, the study aims to examine how risk factors influence model predictions under different clinical objectives.

1.2 Overview of Cardiovascular Diseases (CVDs)

Cardiovascular diseases (CVDs) refer to a broad group of disorders affecting the heart and blood vessels. The category of CVDs includes several major subtypes, such as coronary artery disease, cerebrovascular disease (including stroke), heart failure, arrhythmias, peripheral arterial disease, and hypertensive heart disease (WHO, 2025).

CVDs are influenced by various factors, mainly categorized as modifiable and non-modifiable risk factors. Age, sex, and family history are well-established non-modifiable factors (Payne, 2012). Modifiable risk factors include behavioral factors (e.g., tobacco use, alcohol, diet, etc.), metabolic factors (e.g., lipids, blood pressure, diabetes, obesity), socioeconomic and psychosocial factors (e.g., education, symptoms of depression), etc. (Yusuf, et al., 2020).

1.3 Importance of CVD Risk Prediction

The global burden of CVDs keeps on rising even though medical treatment and public health awareness have improved (Chong, et al., 2025). One main challenge in cardiovascular care is that many serious events, such as myocardial infarction or stroke, happen without any warning beforehand. Importantly, CVD risk does not only appears in people with unhealthy lifestyles, but also in those who look healthy (Kathiresan & Srivastava, 2012).

Traditional risk models such as the Framingham Risk Score (FRS) and ASCVD equations have been widely used for estimating CVD risk. More recently, ML approaches have shown improved predictive performance by modeling complex interactions among risk factors.

Despite these advances, prediction models must not only achieve high performance but also provide transparency that can support clinical decision-making. Therefore, developing accurate and interpretable prediction systems is still an important research challenge.

1.4 Explainable AI in Healthcare Decision Support

Explainable Artificial Intelligence (XAI) aims to make ML models more transparent by providing explanations for their predictions. In healthcare, explainability is particularly important because clinical decisions often require justification and trust.

XAI techniques can help identify the factors that contribute most strongly to model predictions and allow clinicians to better understand model behavior. Consequently, XAI has become an increasingly important component of healthcare decision-support systems, particularly when complex ML models are employed.

1.5 Problem Statement

Although traditional risk scores and ML models are widely used for CVD prediction, several problems remain. Traditional models are easy to understand but may not capture complex relationships between risk factors. Conversely, advanced ML models with higher accuracy are often hard to interpret. Therefore, there is a need for a framework that can both improve prediction performance and provide clear explanations using XAI.

Another difficulty is to distinguish two different predictive tasks in CVD risk study: diagnosis and prognosis. Diagnosis is to identify the patients' illness based on the symptoms determined from clinical measurements at the time, while prognosis addresses the estimation of future disease risk over a certain period of time. Most previous studies mainly focus on improving model accuracy and did not consider comparing these two prediction tasks in detail, especially in terms of feature importance and decision behavior. This problem becomes more obvious when using different datasets, such as the Cleveland Heart Disease dataset and the Framingham dataset, which have different data characteristics. Because of this, relying only on evaluation metrics such as accuracy or F1-score may not be enough.

Therefore, there is a need for an explainable framework that can compare CVD prediction in both diagnostic and prognostic settings, as well as assessing the reliability of model explanations.

1.6 Research Objectives

The primary objective of this study is to investigate how XAI can be used to analyze and compare CVD prediction across different clinical scenarios. Specifically, as mentioned, this research focuses on two distinct problem settings: diagnosis (Cleveland dataset) and prognosis (Framingham dataset).

To reach the aim of developing an explainable ML framework for these different settings, domain-aware data preprocessing pipeline is a must for these clinical datasets, to address clinically meaningful outliers, missing values, imbalance in both features and target variable.

Another objective is to train and evaluate six ML classifiers on both datasets independently, to find the best-performing model for each predictive task based on different evaluation metrics. Next task is to apply SHAP to that best model on each dataset in order to generate global and local post-hoc explanations of model predictions, identifying the most influential features driving CVD diagnosis and ten-year coronary heart disease (CHD) risk prognosis respectively.

Instead of prioritizing predictive performance, this study focuses on understanding how different risk factors influence model predictions. By applying SHAP, an XAI technique, the research aims to identify and compare features contributing in short-term diagnostic prediction and long-term prognosis prediction.

1.7 Scope and Limitations

This study is restricted to two publicly available benchmark datasets: the Cleveland Heart Disease dataset for diagnostic prediction and the shorten Framingham dataset 10-year CHD risk prediction. This study focuses on structured clinical data related to CVD risk factors and does not include imaging data or unstructured clinical data. The analysis is limited to binary risk prediction.

Another limitation is that both datasets are small compared to datasets used in modern large-scale clinical AI research. Moreover, the Framingham dataset exhibits several significant data quality issues, which will be discussed in detail at section 3.3 of Chapter 3.

Six classifiers are evaluated: Logistic Regression, Decision Tree, Support Vector Machine, Random Forest, XGBoost, and CatBoost. Deep learning techniques are not considered.

Explainability is primarily conducted using SHAP, with ALE included as a supplementary method to assess consistency. Explanation reliability is further assessed through stability and agreement analyses. While explainability techniques are employed to enhance interpretability, the study does not aim to replace clinical judgment.

The study does not incorporate external clinical validation, so the key limitation is that the findings are derived from benchmark-specific data distributions. Cross-dataset evaluation is also considered clinically and technically inappropriate for this study, since there are significant differences exist in population demographics, temporal data collection eras, and feature distributions between the two datasets, as well as the key difference in their target variables targets. Therefore, this study does not aim to produce a universally generalized clinical tool but rather to investigate how XAI reveals the shifting importance of risk factors when the clinical endpoint changes from current diagnosis to 10-year prediction.

1.8 Research Methodology Overview

The proposed framework consists of four main stages: data preprocessing, model development, model calibration and threshold optimization, and explainability analysis.

After preprocessing, ML models are trained using nested cross-validation (CV) and hyperparameter optimization. Model calibration and threshold tuning are then performed to improve probability estimation and align predictions with clinical objectives. The final models are evaluated using different performance evaluation metrics.

XAI techniques are then applied to the two best-performing models of two datasets to generate global and local explanations. Finally, consistency and stability analyses are conducted to evaluate the reliability of the generated explanations.

1.9 Thesis Structure

This thesis is organized as follows. Chapter 2 reviews existing literature on CVD risk factors, traditional prediction scores, ML models, and XAI techniques. Chapter 3 presents the dataset, methodology, model development and explanations validation process. Chapter 4 reports experimental results and explainability analyses. Chapter 5 summarizes the findings, discusses limitations and future research directions.

Chapter 2 . LITERATURE REVIEW

2.1 Cardiovascular Disease Risk Factors

2.1.1 Demographic and Lifestyle Factors

Demographic factors are relatively fixed characteristics such as age, sex, ethnicity, and family history that influence the risk of cardiovascular disease (CVD). Age is consistently recognized as one of the strongest predictors, as cardiovascular risk increases with vascular aging and prolonged exposure to risk factors (Payne, 2012; Anderson, Odell, Wilson, & Kannel, 1991). Gender also plays a role, with men generally develop CVD earlier than women.

Lifestyle factors refer to modifiable behaviors such as smoking, diet, physical activity, alcohol consumption, and obesity. Smoking is a major modifiable risk factor and is included in many established risk prediction models (Payne, 2012; Burns, 2003). Obesity and physical inactivity are also associated with increased risk through their effects on blood pressure, lipid metabolism, and glucose regulation (Lavie, Milani, & Ventura, 2009).

The Cleveland and Framingham datasets used in this study contain several demographic and lifestyle variables, including age, sex, smoking status, and indicators related to obesity and physical activity. Other factors such as ethnicity, alcohol consumption, dietary habits, and family history are recognized contributors to CVD risk but are not available in the datasets and are not considered in this research.

2.1.2 Clinical Variables and Biomarkers

Clinical variables are common medical measurements that are widely used in CVD risk prediction models. Common variables include blood pressure, cholesterol levels, and diabetes status, all of which have been shown to be strongly associated with cardiovascular events (Payne, 2012; Lewington, et al., 2002; R Collins, et al., 1990).

These factors are well represented in both datasets used in this study. The Framingham dataset includes systolic and diastolic blood pressure, total cholesterol, diabetes status, and hypertension history. Similarly, the Cleveland dataset contains several clinical indicators related to cardiovascular health, including resting blood pressure, cholesterol level, fasting blood sugar, and exercise-related measurements.

Biomarkers are biological indicators that reflect processes related to cardiovascular disease. Many emerging biomarkers, such as high-sensitivity C-reactive protein (hs-CRP) (Payne, 2012; Emerging Risk Factors Collaboration et al., 2010; Blaha, et al., 2011), apolipoproteins (Ramjee, Sperling, & Jacobson, 2011), troponins (Otsuka, Kawada, Ibuki, & Seino, 2010), and genetic markers (Harismendy, et al., 2011), have also been associated with CVD risk. However, these variables are not available in the Cleveland or Framingham datasets, so they are not the scope of this research.

2.1.3 Imaging and Physiological Signals

Imaging techniques and physiological signals can provide additional information about cardiovascular health in a non-invasive way. Medical imaging and physiological measurements can provide additional information about cardiovascular health. Examples include coronary artery calcium (CAC) scoring, carotid intima-media thickness (CIMT) (Payne, 2012; Bamberg, et al., 2011), flow-mediated dilation (FMD), and pulse wave velocity (PWV) (Payne, 2012; Deanfield, et al., 2005), which have been shown to improve risk assessment in some settings.

However, these measurements require specialized equipment and are not routinely available in many clinical datasets. The datasets Cleveland and Framingham do not contain imaging features or advanced physiological signal data. Therefore, this study focuses only on demographic, lifestyle, and clinical variables.

2.2 Traditional CVD Risk Prediction Models

Before the development of ML approaches, CVD risk assessment relied primarily on statistical risk prediction models. Among the most widely used are the Framingham Risk Score (FRS), the Atherosclerotic Cardiovascular Disease (ASCVD) Pooled Cohort Equations, and the QRISK family of models.

These models estimate an individual's future CVD risk using combinations of traditional clinical factors such as age, sex, blood pressure, cholesterol levels, smoking status, and diabetes. FRS was developed from the Framingham Heart Study and remains one of the most influential risk prediction tools (Wilson, et al., 1998), ASCVD-PCEs were later introduced by the ACC/AHA to estimate 10-year risk of major cardiovascular events, including myocardial infarction and stroke (Goff, et al., 2014). QRISK was developed using United Kingdom population data and incorporates additional factors such as ethnicity, body mass index, and socioeconomic status (Hippisley-Cox J. , et al., 2007).

Although these models are widely used in clinical practice, they rely on predefined statistical relationships and may not fully capture complex interactions among risk factors. As a result, ML approaches have increasingly been investigated as alternatives for improving predictive performance. However, traditional risk models remain important benchmarks and provide the clinical foundation for modern cardiovascular risk prediction research.

Table 2.1 summarizes the characteristics of three models mentioned:

Table 2.1. Traditional CVD Prediction Models

Model	Geographic Focus	Key Predictors	Primary Outcome
Framingham (FRS)	United States	Age, sex, blood pressure, cholesterol, smoking, diabetes	Originally CHD; now General CVD
ASCVD-PCEs	United States	FRS factors + race and treatment status	Hard ASCVD (Heart attack and Stroke)
QRISK (e.g., QRISK3)	United Kingdom	PCE factors + ethnicity, BMI, CKD, social deprivation	Heart attack and Stroke

2.3 Machine Learning Approaches for CVD Prediction

While these conventional approaches above are valuable and well-validated, their predictive performance is limited, especially in heterogeneous populations and when attempting to integrate large-scale or complex data types. Machine Learning (ML) – a part of artificial intelligence (AI) – is increasingly being used in the prediction of CVD over the past few years. ML is widely suggested for heart disease prediction since it can take more efficient and accurate data from large datasets, making prediction simple (Azmi, et al., 2022). This flexibility allows ML models to capture nonlinear interactions and subtle risk signals that may be missed by conventional models. The studies that will be reviewed next will focus on the two datasets that are used in this research – the Framingham dataset and the UCI Cleveland dataset.

Recent study by Sadhukhan et al. (2025) employing the Framingham Heart Study dataset tested a wide range of ML models: decision tree (DT), kNN, Logistic Regression (LR), Naive Bayes, SVM, Random Forest (RF), Gradient Boosting (GB), AdaBoost, a hypertuned GB model, and a StackingCV ensemble. Among these, the StackingCV ensemble achieved the highest performance, with 95% Accuracy, Sensitivity, Specificity, Precision, and F1-score, outperforming individual classifiers such as RF (91%) and hypertuned GB (93%). To address the “black-box” nature of this ensemble method, the authors integrated XAI techniques (SHAP and LIME), to enhance transparency. Despite these strengths, the study acknowledges limitations such as reliance on a single dataset, potential issues of generalizability to diverse populations, and the need for real-world validation before clinical deployment.

Another study using the Framingham dataset was from Suhatri et al. (2024), utilizing multiple classification models including LR, SVM, kNN, RF, etc. RF emerged as the best performing ML model with Accuracy, Precision, Recall, And F1 scores of 0.85, 0.84, and 0.79, respectively. The limitation of this study is that they did not deeply address the imbalance of the dataset, which is a one of the most critical problem.

Another study of Habashneh et al. (2025) using UCI Cleveland heart disease data (Janosi, Steinbrunn, Pfisterer, & Detrano, 1989), presented a comparative analysis of four common ML classifiers – LR, SVM, kNN and RF. kNN had the highest Accuracy of 0.908, while RF is the best in Recall (0.886). The study emphasized the trade-off between

Accuracy with clinical safety, but in healthcare setting, models that are better at catching sick patients (high Recall) like RF is more suitable.

A recent work of Umar et al. (2025) proposed an XAI framework for CVD prediction, integrating SHAP and Permutation Feature Importance (PFI) into ML models to enhance interpretability. Although their broader framework incorporates Internet of Things (IoT) components for real-time monitoring, the core contribution relevant to this work lies in their use of SHAP to provide patient-level and population-level explanations for model predictions. They evaluated multiple models: RF, SVM, kNN, and Deep Neural Network (DNN) across three datasets: two real-world (Cleveland (Janosi, Steinbrunn, Pfisterer, & Detrano, 1989) and IEEE combined dataset (Siddhartha, 2020)) and one synthetic; highlighting the importance of physiological features in improving model accuracy. RF had the highest accuracy on real and synthetic datasets (up to 98.06%). SHAP visualizations effectively illustrated feature contributions to individual predictions. This study offers a valuable benchmark for comparing both predictive performance and explainability outcomes in CVD risk modeling.

There are several studies that considered using both Framingham and Cleveland datasets for their prediction of CVD. Wu et al. (2024) proposed novel stacking model called SDKABL, combining SVM, DT, and kNN at the base layer and the ABiLSTM model at the meta layer. Besides two datasets Framingham and Cleveland, the study also used Z-Alizadeh Sani dataset. Across all datasets, SDKABL consistently outperformed individual classifiers, achieving Accuracies above 90% with notable gains in recall and precision when PCA was incorporated. The highlight of this study is the synergy between PCA feature fusion and the ABiLSTM meta-layer, which significantly boosted predictive stability and generalization.

Adekoya et al. (2025) also introduced an interpretable framework which combines Explainable Boosting Machine (EBM) with XGBoost, SHAP and LIME, called Interpretable Ensemble Learning Framework (IELF). Using the two benchmark datasets, IELF achieved strong results on Cleveland, with $AUC \approx 96\%$, $Accuracy \approx 88.5\%$, and $Recall \approx 92.9\%$, matching or slightly outperforming the base models. On Framingham, IELF balanced the trade-offs between EBM (high Accuracy but very low Recall) and XGBoost (high recall but poor accuracy), reaching $Accuracy \approx 82.6\%$, $Recall \approx 16.3\%$, and the best $AUC \approx 68.6\%$.

Table 2.2 provides the summary of all reviewed studies:

Table 2.2. Summary of Reviewed Studies on ML for CVD Prediction

Study	Data Used	ML Models / Methods	Key Results	Limitations / Challenges
Sadhukhan et al. (2025)	Framingham	10 ML models, including StackingCV classifier + XAI (SHAP and LIME)	StackingCV ensemble – highest performance: 95% Accuracy, Sensitivity, Specificity, Precision, and F1-score	Reliance on a single dataset; real-world validation
Suhatril et al. (2024)	Framingham	DT, SVM, kNN, LR, RF, GB	RF best ($\approx 84\%$ Accuracy)	Imbalance handling limited
Habashneh et al. (2025)	UCI Cleveland	LR, SVM, kNN and RF	kNN – highest Accuracy (90.8%) RF – highest Recall (88.6%)	Trade-off between Accuracy and clinical safety
Umar et al. (2025)	2 real-world and 1 synthetic CVD dataset (IEEE + UCI Cleveland + synthetic)	RF, SVM, kNN, DNN + XAI (SHAP)	RF achieved highest accuracy (up to $\approx 98\%$)	Reliance on synthetic data may limit generalizability and clinical reliability without real-world validation
Wu et al. (2024)	3 datasets: Framingham + Cleveland + Z-Alizadeh Sani dataset	stacking model SDKABL	SDKABL Accuracy $\approx 90\%$	Low interpretability

Adekoya et al. (2025)	2 datasets: Framingham and Cleveland	ensemble + XAI (IELF)	Framingham - Accuracy $\approx$ 82.6%, Recall $\approx$ 16.3%, and the best AUC $\approx$ 68.6% Cleveland - AUC $\approx$ 96%, Accuracy $\approx$ 88.5%, and Recall $\approx$ 92.9%	Dataset heterogeneity
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2.4 Explainable AI (XAI) Techniques

This section provides further information about XAI techniques based on how they are generally categorized.

2.4.1 Interpretable Models (Logistic Regressions, Decision Tree)

Interpretable models provide explanations directly through their model structure. Logistic Regression (LR) explains predictions using feature coefficients, where the magnitude and sign of each coefficient indicate its influence on the outcome. Decision Trees (DT) are also inherently interpretable because predictions are generated through a sequence of transparent decision rules.

Although interpretable models are easy to understand, they may not capture complex nonlinear relationships as effectively as more advanced ML algorithms. This trade-off between interpretability and predictive performance has motivated the use of post-hoc explanation methods for complex models.

2.4.2 Post-hoc Explainability Methods

Post-hoc methods explain model predictions after training and can be applied to complex ML models. Among these methods, SHAP has become one of the most widely used techniques because it provides both local and global explanations based on cooperative game theory (Lundberg & Lee, 2017). SHAP values quantify the contribution of each feature to a prediction by computing its marginal contribution across all possible feature subsets. As just mentioned, SHAP can be applied in both local and global contexts: local explanations interpret individual predictions, while aggregated SHAP values reveal overall feature importance patterns.

Other than SHAP, Accumulated Local Effects (ALE) plots are commonly used to visualize how features influence model predictions while reducing the feature-correlation issues associated with Partial Dependence Plots (PDPs). In this study, SHAP is used to analyze feature importance and individual predictions, while ALE plots are for supporting explanation consistency analysis.

2.4.3 Model-Agnostic vs. Model-Specific Methods

XAI methods can be categorized as model-agnostic or model-specific. Model-agnostic methods generate explanations without relying on the internal structure of a model and can therefore be applied consistently across different algorithms. Examples include SHAP and ALE.

Model-specific methods utilize information from a particular model type, such as coefficients in LR or feature importance measures in tree-based models. While these methods are computationally efficient, their applicability is limited to specific algorithms.

2.5 Challenges in Applying XAI to Healthcare

Applying explainable AI to healthcare encounters profound challenges that span technical, ethical, and practical dimensions, creating barriers that extend far beyond simple algorithmic implementation.

2.5.1 Bias and Fairness Issues

One of the foremost challenges in deploying XAI in healthcare is addressing bias and ensuring fairness in both the underlying models and the explanations they produce. Bias can originate at multiple stages of AI development, including data collection, annotation, model training, and evaluation (Mienye, et al., 2024). Healthcare data often reflects real-world demographics that are unevenly represented (for example, certain age groups, ethnicities, or socio-economic statuses), which can result in systems that perform differently across subpopulations. When such biased systems are used in clinical decision support, they can exacerbate disparities in diagnosis, treatment recommendations, and patient outcomes if not carefully mitigated and audited. XAI methods are intended to make models more transparent, but they do not inherently guarantee that the explanations themselves are free of bias, some explanation techniques may inadvertently hide or even amplify inequities if not rigorously evaluated across diverse patient groups.

2.5.2 Clinical Interpretability

Healthcare professionals require explanations that are understandable and consistent with clinical knowledge (Sadeghi, et al., 2024). . Although modern ML models often achieve high predictive performance, their decision-making processes are frequently difficult to interpret. XAI techniques help bridge this gap by providing insights into the factors influencing predictions, thereby improving transparency and supporting clinical decision-making.

2.5.3 Data Quality Constraints

The quality of healthcare data directly affects both predictive performance and explanation reliability. Missing values, class imbalance, measurement errors, and limited sample sizes can influence model behavior and potentially lead to misleading explanations. These issues show the importance of careful data preprocessing and robust model evaluation procedures.

2.6 Research Gaps and Summary

This chapter reviewed a wide range of studies related to CVD risk factors, traditional risk prediction models, ML approaches, and XAI techniques.

Traditional cardiovascular risk prediction models such as Framingham Risk Score (FRS), ASCVD-PCEs, and QRISK remain widely used because of their simplicity and clinical interpretability. However, these models are based on predefined statistical relationships and may not adequately capture complex interactions among risk factors.

ML approaches have demonstrated superior predictive performance in many CVD prediction tasks by modeling nonlinear relationships and interactions more effectively. However, most studies focus primarily on improving predictive accuracy, not on understanding how models come up with their predictions.

As model complexity increases, interpretability becomes more important. Recent studies have incorporated XAI techniques, particularly SHAP, to explain model predictions. However, most existing work just used explanation visualization but did not evaluate the reliability of those explanations. The stability of explanations across different training samples and the consistency between different explanation techniques are rarely assessed.

Another limitation is the lack of distinction between different CVD prediction scenarios. Diagnostic tasks, which identify existing disease, and prognostic tasks, which estimate future cardiovascular risk, are really different in terms of objectives, data characteristics, and clinical applications. Despite these differences, many studies treat them as a single prediction problem.

Therefore, this study addresses three key gaps. First, it compares ML models across two distinct CVD prediction scenarios: diagnosis and prognosis. Second, it applies XAI techniques to provide both global and local explanations of model behavior. Third, it evaluates the reliability of explanations through consistency analysis using SHAP and ALE, and stability analysis using bootstrap SHAP, feature-ranking agreement, and overlap-based measures. These analyses aim to provide more trustworthy and clinically meaningful explanations for CVD prediction.

Chapter 3 . METHODOLOGY AND DATA DESCRIPTION

3.1 Research Framework

The framework consists of five major stages: data preparation, model selection, model refinement, final evaluation, and explainability analysis.

Stage 1: Dataset Preparation

Both datasets, particularly Framingham, will first go through data cleaning and reformatting, where the numerically preprocessed categorical columns are reformatted into their original data types, and extremely clinically unreasonable values are clipped to the designated ranges, the ranges are all based on official clinical sources and possible extreme cases recorded. The Cleveland dataset only needs rearranging the target variable to be binary instead of multiclass.

Each dataset is initially divided into training (80%) and testing (20%) subsets using stratified sampling to preserve the original class distribution. The test set is reserved exclusively for final evaluation and is not used during model development.

Stage 2: Model Selection Using Nested CV

Now all preprocessing, resampling, and model tuning steps are performed exclusively on the training data only. The train set will be preprocessed accordingly to each model category: linear (LR), distanced-based (SVM), tree-based (DT, RF, XGB, CB); including handling missing values, feature transformation, scaling, and encoding where required.

Based on the mean $\pm$ standard deviation performance across the outer folds, the three best-performing models are selected for further analysis. Model selection criteria differ according to the prediction task. ROC-AUC is used for the Cleveland dataset, whereas PR-AUC is used for the Framingham dataset due to class imbalance.

Stage 3: Model Refinement

After model selection, hyperparameter tuning is performed on the whole training set to gain the best configuration for each selected model. The hypertuned models are then calibrated using probability calibration techniques to improve probability estimates.

Following calibration is the threshold tuning stage, using out-of-fold predicted probabilities generated through cross-validation. For the Cleveland dataset, the optimal threshold is determined by maximizing Youden's J statistic on the ROC-curve. For the Framingham dataset, the optimal threshold is determined by maximizing the F2-score on the Precision-Recall curve. The optimal threshold is then fixed and contained into the final classifier.

Stage 4: Final Model Evaluation

The calibrated and threshold-optimized models are evaluated on the independent test set using multiple evaluation metrics. The best-performing model for each dataset is identified during this stage and selected for explainability analysis.

Stage 5: SHAP Analysis

The final stage applies XAI technique – SHAP, to interpret model predictions. SHAP is used to generate both global and local explanations to show the most influential risk factors and the contribution of individual features to specific predictions.

Some additional validation analyses are performed to confirm SHAP explanations reliability. ALE plots are used to compare feature-effect patterns with SHAP dependence plots, Bootstrap SHAP analysis and ranking agreement measures are employed to evaluate explanation stability and consistency.

The whole framework is simply illustrated in Figure 3.1 below.

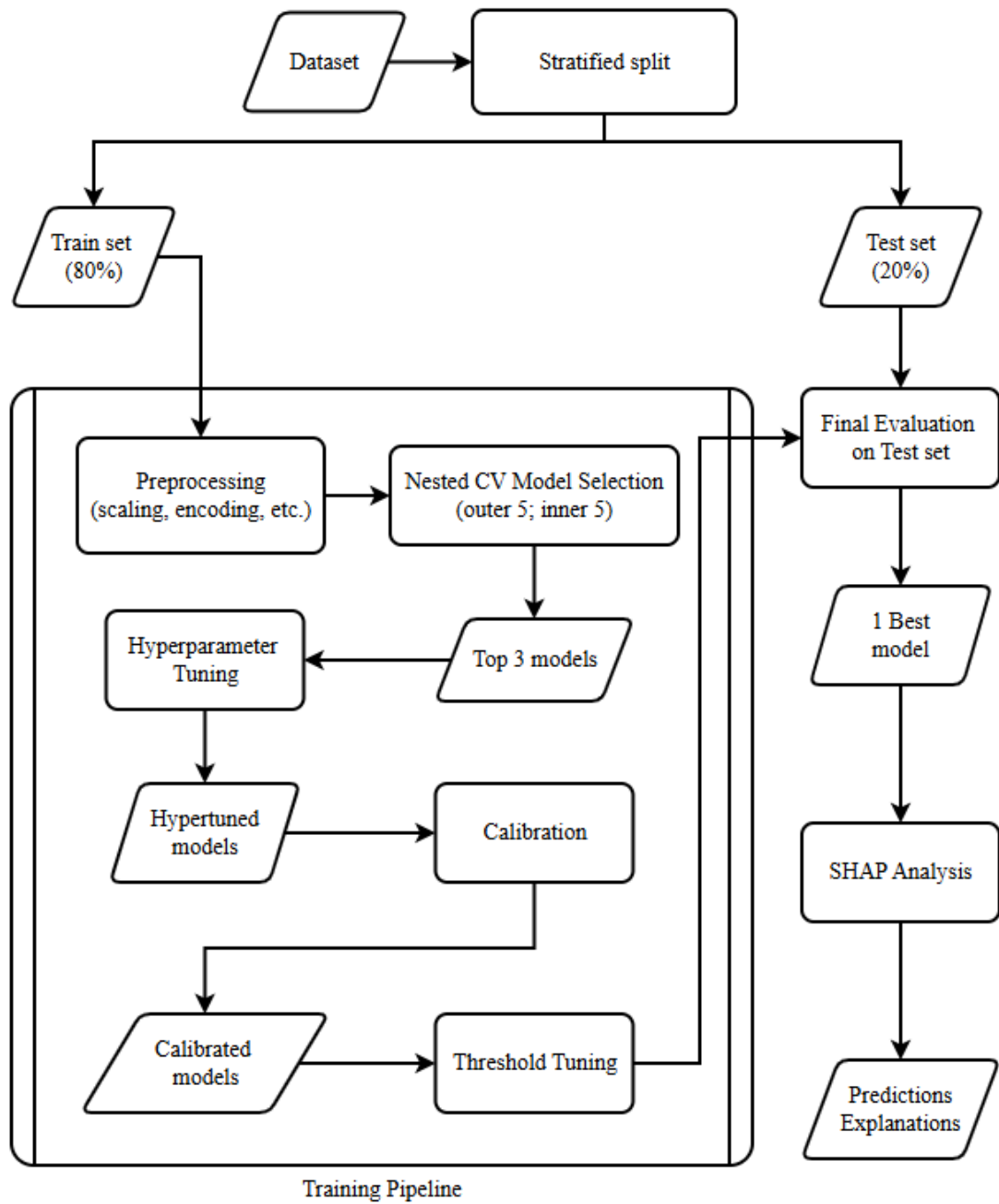


Figure 3.1. Research Framework

3.2 Dataset Description

3.2.1 Data Sources

One of the most crucial step of developing a CVD prediction model is data collection. There are many clinical data sources publicly available for this specific type of diseases. The chosen datasets are 2 of the most well-known real-world datasets that used especially

for research: the Cleveland dataset of the “Heart Disease” Database from the University of California Irvine (UCI) ML Repository (Janosi, Steinbrunn, Pfisterer, & Detrano, 1989), with 303 instances and 14 attributes; and the short-version dataset of the Framingham Heart Study obtained from Kaggle (Bhardwaj, 2022) with over 4240 instances and 16 attributes. These datasets contain patients demographic and health metrics features, such as age, gender, cholesterol, heart rate, blood pressure, etc., which are all CVD risk factors that should be taken into consideration.

The description for the datasets will be discussed in detail in the next section (Section 3.2.2).

3.2.2 Variables Definitions and Clinical Significance

As mentioned, the first dataset – the UCI Cleveland Heart Disease dataset, includes 303 instances 14 attributes, 13 of which are input variables, the rest is the binary target variable (target) (Table 3.1).

Table 3.1. The UCI Heart Disease Dataset Description

Attribute	Description	Data Type
<i>Clinical variables</i>		
age	Age in years	Numerical
sex	Gender 1 = male 0 = female	Binary
cp	Chest pain type 1 = typical angina 2 = atypical angina 3 = non-anginal pain 4 = asymptomatic	Nominal
trestbps	Resting blood pressure (mm/Hg) (systolic blood pressure)	Numerical
<i>Routine test data collected</i>		
chol	Serum cholesterol (mg/dl)	Numerical
fbs	Fasting blood sugar > 120 mg/dl 1 = true 0 = false	Binary

restecg	Resting electrocardiogram results 0 = normal 1 = having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) 2 = showing probable or definite left ventricular hypertrophy by Estes' criteria	Nominal
<i>Noninvasive tests data collected</i>		
thalach	Maximum heart rate achieved	Numerical
exang	Exercise induced angina 1 = yes 0 = no	Binary
oldpeak	ST depression induced by exercise relative to rest	Numerical
slope	The slope of the peak exercise ST segment 1 = upsloping 2 = flat 3 = downsloping	Ordinal
ca	Number of major vessels (0-3) colored by fluoroscopy	Numerical
thal	Exercise thallium scintigraphic defects 3 = normal 6 = fixed defect 7 = reversible defect	Numerical
<i>Target Variable</i>		
num	Diagnosis of heart disease (angiographic disease status) 0 = $<50\%$ diameter narrowing 1 = $>50\%$ diameter narrowing	Binary

To make it clearer, the variables are divided as clinical variables and collected data from routine test and noninvasive test. Clinical variables include age, gender, chest pain type and systolic blood pressure; routine test data consists of serum cholesterol, fasting blood sugar >120 mg/dl and electrocardiographic results at rest. There are three noninvasive tests: exercise electrocardiogram, exercise thallium scintigraphy and fluoroscopy for coronary calcium; the exercise and fluoroscopic data includes all other variables left except for the target variable.

The target variable classifies those whose coronary angiograms were considered abnormal (value 1) as positive cases of CVD, and vice versa.

As for the second dataset – Framingham, there are 15 features and a binary dependent variable, with 4240 instances. This dataset was originally from the Framingham Heart Study dataset, which contains information patients participated in a CVD study in

Framingham, Massachusetts. This shortened version is a valuable and well-known data source that researchers and data specialists prefer for their studies on healthcare modeling.

Table 3.2 clearly shows the categories separation of the features. Patients demographics include gender, age and education levels; smoking habits, blood pressure, cholesterol levels, BMI and glucose levels are categorized as health metrics; medical history involves examining whether the patients have ever had stroke, hypertension, diabetes and be on blood pressure medication.

The target variable considers those who are at risk of CHD in 10 years as 1 (yes) and those who are not as 0 (no).

Table 3.2. The Framingham Heart Study Dataset Description

Attribute	Description	Data Type
<i>Demographics</i>		
male	Gender 1 = Men 0 = Women	Binary
age	Age of the patient in years	Numerical
education	Education level of the patient (1 to 4). 1 = 0-11 years 2 = High School Diploma, GED 3 = Some College, Vocational School 4 = College (BS, BA) degree or more	Ordinal
<i>Health Metrics</i>		
currentSmoker	Indicates if the patient is a current smoker (1) or not (0)	Binary
cigsPerDay	Number of cigarettes smoked per day	Numerical
totChol	Total cholesterol level of the patient	Numerical
sysBP	Systolic blood pressure of the patient	Numerical
diaBP	Diastolic blood pressure of the patient	Numerical
BMI	Body Mass Index of the patient	Numerical
heartRate	Heart rate of the patient	Numerical
glucose	Glucose level of the patient	Numerical
<i>Medical History</i>		
prevalentStroke	Indicates if the patient has a history of stroke (1) or not (0).	Binary

prevalentHyp	Indicates if the patient has a history of hypertension (1) or not (0).	Binary
diabetes	Indicates if the patient has diabetes (1) or not (0).	Binary
BPMeds	Indicates if the patient is on blood pressure medication (1) or not (0).	Binary
Target Variable		
TenYearCHD	Indicates the 10-year risk of CHD (1) or not (0).	Binary

3.2.3 Data Cleaning

Data cleaning is the very necessary early stage in the data analysis process. The goal of data cleaning is to fix, or discard incorrect, duplicated or incomplete data accordingly the result received from Exploratory Data Analysis (EDA) process.

The first step of data cleaning was to categorize the features in their corresponding data types (numerical, binary, nominal, ordinal, etc.) for convenient preprocessing (as showed in section 3.2.2). EDA was later conducted based on the data types (numerical or categorical). The preprocessing process will be decided accordingly to EDA findings and assumptions.

After EDA, a step of clipping/winsorizing has been implemented, particularly for the Framingham dataset where some of the outliers are clinically unreasonable and unrealistic. The boundaries are defined in Table 3.3 below.

For the Cleveland dataset, since the focused task in this study is binary classification, all other classes higher than “1”, which also imply diagnosis of disease, have been changed to “1”.

Table 3.3. Clipping boundaries for some clinical features of Framingham dataset

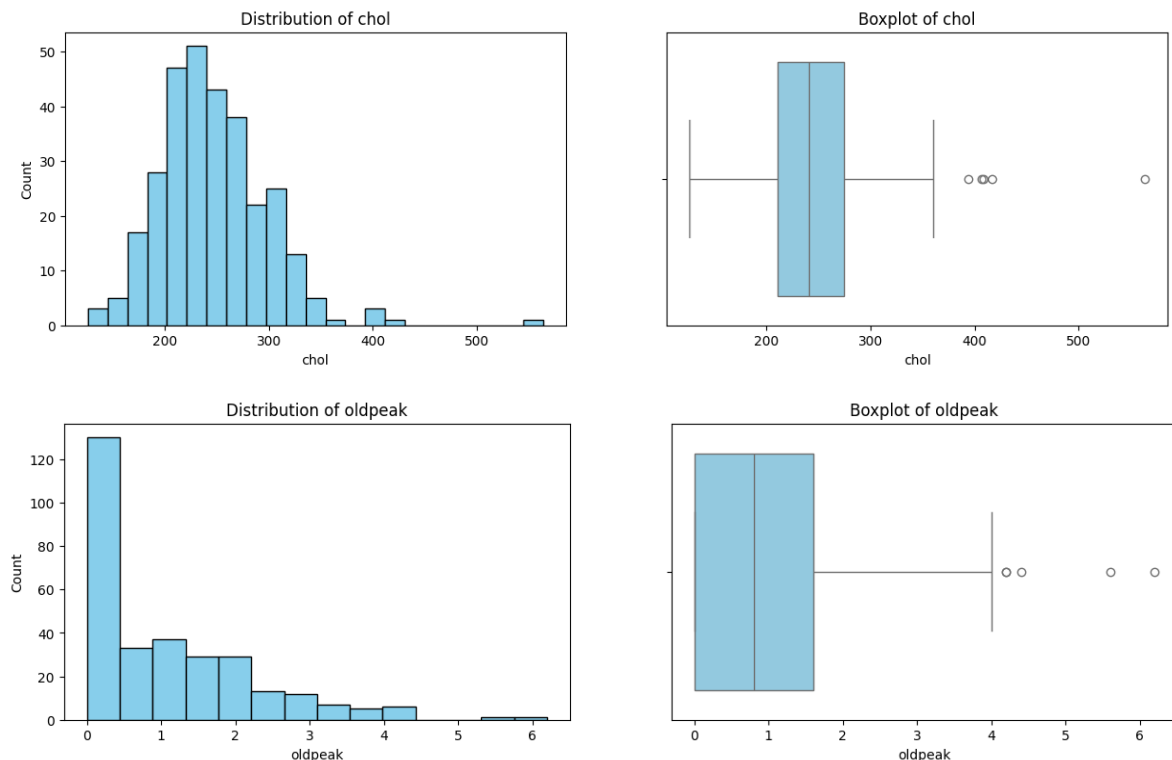
Features	Upper bound	Lower bound
age	20	80
cigsPerDay	0	60
totChol	100	600
sysBP	70	250
diaBP	40	150
BMI	10	60

heartRate	30	200
glucose	40	500

3.3 Exploratory Data Analysis (EDA)

Later ML tasks require deep understanding of the data to handling it efficiently. Therefore, EDA was performed as a separate initial step to understand the basic structure and clinical patterns of the cardiovascular dataset without modifying the data. EDA shows that both datasets have no duplicate rows, but all have missing values. These missing values are decided to be retained instead of deleted, the imputation method will be discussed further in the next section.

From manual visualizations, the study analyzed feature distributions and found that many numerical variables were skewed and included notable outliers (“chol” and “oldpeak” of Cleveland, “cigsPerDay” and “glucose” of Framingham, are most skewed ones), which is typical in clinical data (Figure 3.2).



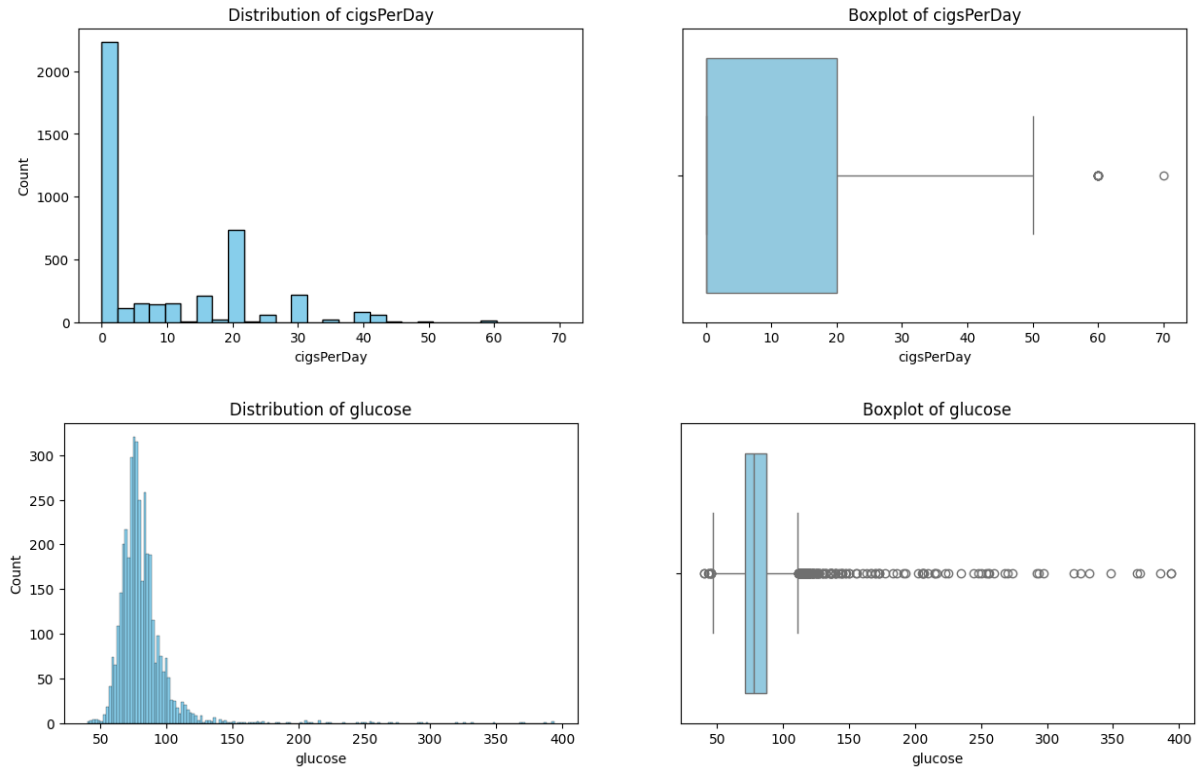


Figure 3.2. Histograms and Boxplots of “chol” and “oldpeak” (Cleveland) (upper 2); “cigsPerDay” and “glucose” (Framingham) (lower 2)

To compare distributions for numerical features for different values of “target”, KDE plots were conducted. The KDE plots indicated that age and blood pressure have remarkably distinct distributions when comparing patients who may have CHD in 10 years, versus those who may not (Figure 3.3). While in the case of diagnosis, age, ST depression induced by exercise relative to rest (oldpeak), number of major vessels colored by fluoroscopy (ca) and maximum heart rate achieved (thalach) have high possibility to be the most significant factors that influence the prediction (Figure 3.4).

The EDA also showed data sparsity and uneven distribution in categorical variables, where some categories occurred much more often than others, despite low cardinality (four categories/feature or fewer). Target imbalance problem was also spotted in the Framingham dataset, with only about 15% of the data is positive class.

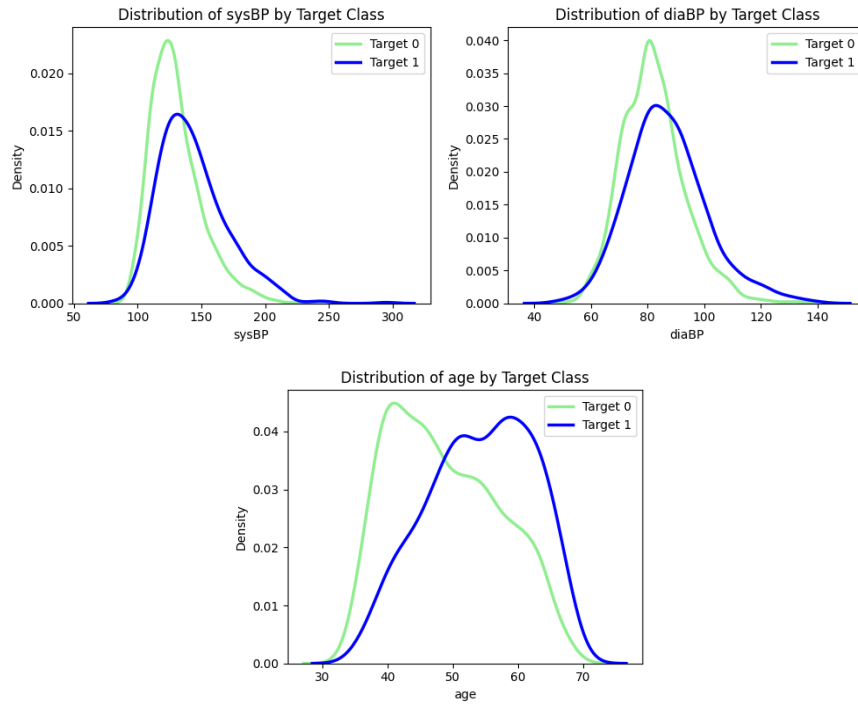


Figure 3.3. KDE plots of some numerical features of Framingham dataset

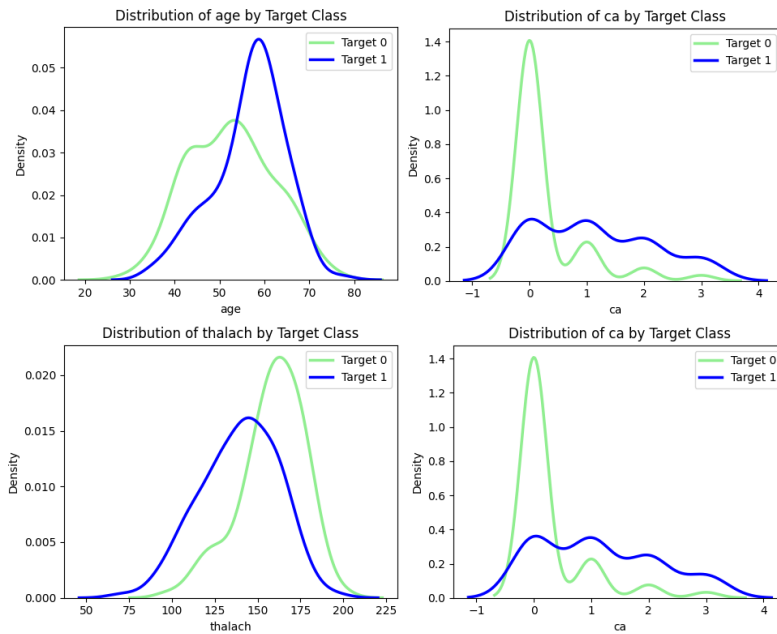


Figure 3.4. KDE plots of some numerical features of Cleveland dataset

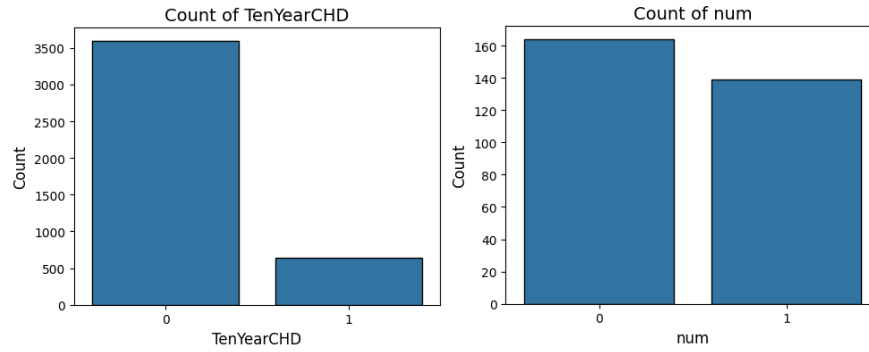


Figure 3.5. Bar plots of the target variables: *TenYearCHD* of the Framingham dataset and *num* of the Cleveland dataset.

The correlation matrices (Figure 3.6) showed that in the Cleveland dataset, no features are significantly correlated. However, the Framingham dataset is the opposite, with many features strongly correlated, especially those closely related and directly connected medically, such as “sysBP” and “diaBP”, “diabetes” and “glucose”, “cigsPerDay” and “currentSmoker”, etc.

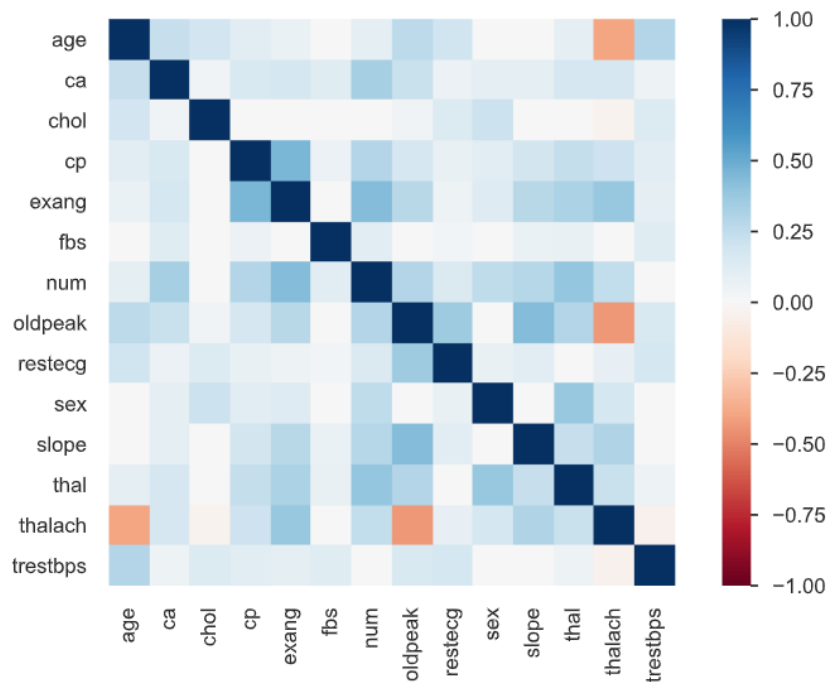


Figure 3.6. Correlation matrix of Cleveland dataset

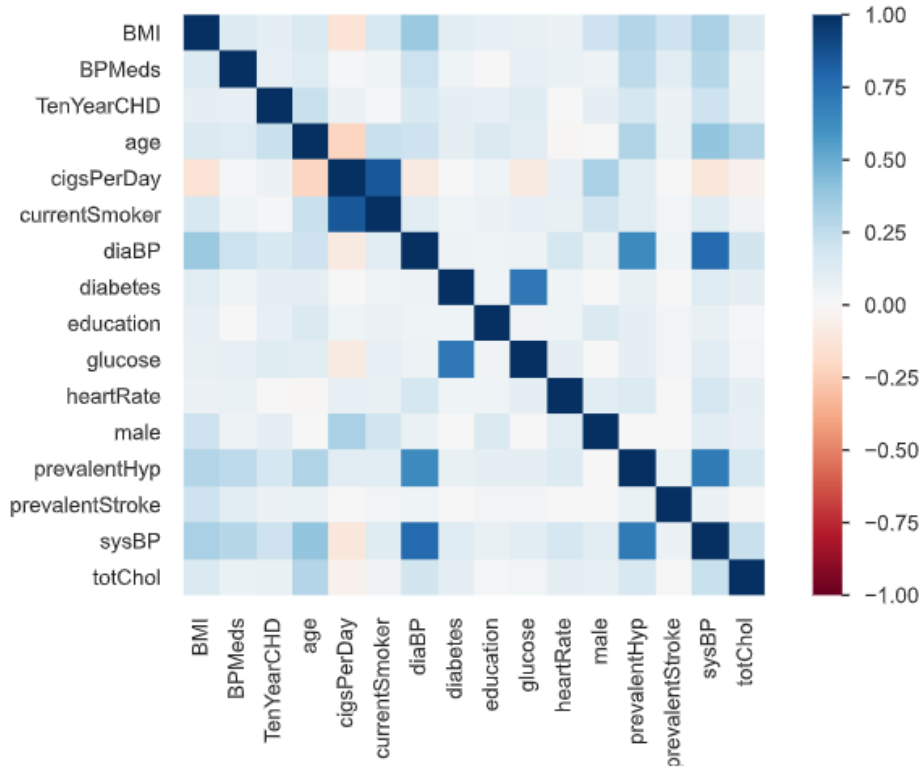


Figure 3.7. Correlation matrix of Framingham dataset

3.4 Data Preprocessing

This study employs a model-specific preprocessing strategy.

Starting with missing value handling, instead of using simple mean or median imputation, which can distort data distribution and reduce variance, missing values were handled using Multivariate Imputation by Chained Equations MICE for numerical columns. This technique treats each feature with missing data as a target variable and models it as a function of other variables in the dataset.

To reduce skewness and improve distributional properties, Yeo-Johnson transformation was applied to numerical features for distance-based models. Robust Scaling was subsequently used to minimize the influence of outliers commonly found in clinical data. Categorical variables were encoded using different techniques depending on the model family: Weight of Evidence (WoE) encoding for LR, James-Stein encoding for distance-based models SVM, and Ordinal Encoding for tree-based models. Table 3.4 summarizes the preprocessing techniques applied in this study.

Table 3.4. Data Preprocessing

Preprocessing Step	Technique	Purpose	Applied Models
Missing Value Handling	MICE, mode	Estimate missing numerical values using multivariate relationships	All models
Distribution Transformation	Yeo-Johnson Transformation	Reduce skewness and improve feature distributions	SVM
Feature Scaling	Robust Scaling	Reduce sensitivity to outliers using median and IQR	LR, SVM
Categorical Encoding	Weight of Evidence (WoE)	Transform categories into target-related numerical values	LR
Categorical Encoding	James-Stein Encoding	Reduce overfitting through shrinkage-based target encoding	SVM
Categorical Encoding	Ordinal Encoding	Change back the categories to numeric format	DT, RF, XGB, CB

3.5 Model Development

3.5.1 Baseline Statistical Models

Logistic Regression (LR) is chosen as baselines due to its inherent interpretability through coefficients.

Logistic Regression – LR estimates the probability that a patient is of the positive class (heart disease or high 10-year CHD risk), using the logistic function (sigmoid curve) to map a linear combination of input clinical features to a value between 0 and 1. In medical research, LR is the industry standard due to its high interpretability.

3.5.2 Machine Learning Models

To capture non-linear relationships that statistical baselines might miss, two ML algorithms were implemented.

Decision Tree (DT) – DT is a rule-based model. At each node, the model chooses the feature that best separates patients into different outcome groups. For example, the model may first split patients according to chest pain type, then further divide them based on other features, until reaching a final prediction.

Support Vector Machines (SVM) – works by finding the best boundary that separates positive and negative patients. SVM can learn nonlinear relationships by using a kernel function. For example, patients with same blood pressure may still have different risks depending on their age and cholesterol level.

3.5.3 Ensemble Models

Advanced ensemble methods were employed to address the limitations of single models.

Random Forest (RF) – trains multiple DT on different random subsets of training data and features, then averages their predictions. For example, if most trees predict that a patient has heart disease, the final prediction will also be positive. This reduces variance and is highly robust to noise in clinical data.

Extreme Gradient Boosting (XGBoost/XGB) – builds trees sequentially. Each new tree tries to correct the errors (residuals) of the previous ones. XGBoost is chosen for its execution speed and ability to handle missing data and complex patterns effectively.

Categorical Boosting (CatBoost/CB) – same as XGB, but is specifically designed to handle categorical variables more effectively and reduce prediction bias during training. This is why it is particularly suitable for datasets with many categorical variables like Cleveland and Framingham.

For the Framingham dataset, imbalance was handled using algorithm-level techniques. Class weighting was applied for all other the models and scale-adjusted class weighting was applied for XGB, while decision threshold tuning was performed after probability calibration, to further align predictions with the clinical objectives of each prediction task.

3.6 Explainability Modules

3.6.1 SHAP-based Global and Local Explanations

To improve the interpretability of ML models, this study employs SHapley Additive exPlanations (SHAP), a game-theoretic explainability framework proposed by Lundberg and Lee (2017). SHAP quantifies the contribution of each feature to a prediction by estimating its marginal contribution across different feature combinations.

SHAP supports both global and local explanations. Global explanations were generated using SHAP Beeswarm and SHAP Bar plots to find the most influential CVD risk factors across the dataset. Local explanations were generated using Waterfall and Force plots to explain one single patient predictions.

Since the focus of this study is application rather than theoretical derivation, only the core SHAP concepts are presented. The detailed mathematical formulations can be found in the original work of Lundberg and Lee (2017).

3.6.2 Accumulated Local Effect (ALE)

Accumulated Local Effects (ALE) plots were used to show how continuous features affect the model prediction. Unlike Partial Dependence Plots (PDPs), ALE is less affected by highly correlated features because it only uses data that actually exist in the dataset.

In this study, ALE was mainly used as a validation tool. Each ALE plot was compared with the corresponding SHAP dependence plot. If both plots show a similar trend and direction, it indicates that SHAP provides consistent explanations because another independent method produces similar results. This is a cross-method consistency.

3.6.3 Bootstrap Stability

While ALE evaluates cross-method consistency, bootstrap analysis evaluates explanation stability. The test dataset was randomly resampled 20 times, and SHAP values were calculated for each sample.

After each run, the feature importance ranking was recorded. The stability of the explanations was then evaluated using the mean SHAP importance, Kendall's Tau (τ), and Top-K overlap. If these values remain similar across different bootstrap samples, the explanations can be considered stable and reliable.

3.6.4 Visualization of Risk Contributions

As mentioned, SHAP provides multiple visualizations for both global and local explanations. Local explanations have Waterfall and Force plots, while global explanations have Beeswarm and Bar plots. Moreover, ALE plots of global-level explainability and Bootstrap SHAP Feature Ranking plots are also executed for explanations' reliability validation.

3.7 Evaluation Strategy

3.7.1 Predictive Performance Metrics

Model performance was evaluated using Sensitivity, Specificity, Precision, F1-score, Balanced Accuracy, ROC-AUC, and PR-AUC.

Because the two datasets represent different clinical scenarios, different primary metrics were adopted during model selection. For the Cleveland dataset, ROC-AUC was selected as the primary metric because the target classes are relatively balanced and the objective is diagnostic discrimination. For the Framingham dataset, PR-AUC was selected because the dataset exhibits substantial class imbalance, making precision-recall performance more informative than ROC-AUC.

These metrics are primarily derived from the Confusion Matrix, a tabular representation used to display the performance of a classification model on a set of data for which the true values are known.

Confusion Matrix – this matrix (Table 3.5) categorizes every prediction into one of four outcomes:

Table 3.5. Confusion Matrix

	Predicted Negative	Predicted Positive
Actual Negative	True Negative (TN)	False Positive (FP)
Actual Positive	False Negative (FN)	True Positive (TP)

Precision – Precision is calculated by taking the number of correctly predicted positive instances divide by the total of all predicted positive instances.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

Sensitivity (Recall) – This metric measures the model’s ability to find all the actual positive cases in the dataset.

$$Sensitivity = \frac{TP}{TP + FN} \quad (2)$$

In CVD prediction, sensitivity is arguably the most critical metric. High sensitivity ensures that fewer high-risk patients are missed (low False Negatives), allowing for early intervention to prevent fatalities.

F1-Score – This metric is the harmonic mean of Precision and Sensitivity. It provides a single score that balances both metrics.

$$F1 = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity} \quad (3)$$

A high F1-score indicates that the model is performing well at both identifying at-risk patients and keeping false alarms low.

Specificity – This metrics calculates the proportion of true negatives among all actual negatives.

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

Balanced Accuracy – Unlike the usual Accuracy metric, this metric calculates the average of sensitivity (recall) and specificity.

$$Balanced\ Accuracy = \frac{Specificity + Sensitivity}{2} \quad (5)$$

AUC-ROC Score – The Area Under the Receiver Operating Characteristic Curve shows how much the model is capable of distinguishing between classes (e.g., diseased

vs. healthy). An AUC of 1.0 represents a perfect model, while 0.5 suggests the model is no better than random guessing.

PR-AUC Score – The Area Under the Precision-Recall Curve is the average precision across recall levels. ROC-AUC can be misleading especially in the case of imbalance dataset like Framingham.

3.7.2 Calibration Analysis

Probability calibration was performed using CalibratedClassifierCV with Platt Scaling (Sigmoid Calibration) for the Cleveland dataset and Isotonic Calibration for Framingham dataset. Calibration quality was evaluated through calibration curves and Brier scores, which measure the agreement between predicted probabilities and observed outcomes.

3.7.3 Decision Threshold Tuning

After calibration, decision threshold tuning stage was performed using out-of-fold predicted probabilities generated from the training set to avoid data leakage.

Because the two datasets have different clinical objectives, different threshold tuning strategies were applied.

For the Cleveland dataset, the optimal threshold was selected by maximizing Youden's J statistic on the ROC curve:

$$J = \text{Sensitivity} + \text{Specificity} - 1 \quad (6)$$

This approach finds the threshold that shows the best balance between Sensitivity and Specificity for diagnostic prediction.

For the Framingham dataset, the optimal threshold was selected by maximizing the F2-score on the Precision-Recall curve. Compared to the F1-score, the F2-score emphasizes more on Sensitivity.

3.8 Tools and Technologies

All the development, training, and evaluation of the ML models in this research were conducted using Python. The following tools and libraries were selected for their robustness, flexibility, and widespread support.

Google Colab – An interactive, cloud-based Jupyter Notebook environment was used for model development. Colab was chosen for its collaborative features and its provision of high-performance hardware (GPUs and TPUs).

Visual Studio Code (VSCode) – For local script management, advanced debugging, and version control integration, VSCode served as the primary Integrated Development Environment (IDE).

Core ML Libraries:

ydata_profiling: Great help in EDA. Used to generate a report on the data overall structure and simple univariate and multivariate analysis.

scikit-learn: Used for implementing traditional ML algorithms, data splitting, cross-validation, and performance metrics calculation.

category_encoders: Utilized for advanced transformations like James-Stein and Weight of Evidence.

XGBoost / CatBoost: Utilized for ensemble models XGBoost and CatBoost.

pandas and numpy: Used for data manipulation, cleaning, and preprocessing tasks.

matplotlib and seaborn: Used to generate clinical visualizations, including the Confusion Matrix and the Receiver Operating Characteristic (ROC) curves used to determine the AUC score.

SHAP: Integrated to provide global and local explanations of the models.

Chapter 4 . EXPERIMENTS AND RESULTS

4.1 Experimental Setup

As mentioned, the experiments in this study focus on two datasets, which are Framingham dataset (for prognosis task) and the Cleveland dataset (for diagnosis task). The main goal is to compare model performance and later apply XAI methods to understand feature importance in both scenarios.

The Cleveland and Framingham datasets were divided into training and testing subsets using an 80:20 stratified split. Nested CV with five outer folds and five inner folds was used for model selection and hyperparameter optimization. Model calibration was performed using CalibratedClassifierCV with Sigmoid and Isotonic methods depending on the datasets' objectives. Threshold tuning was executed after calibration, applying ROC curve and PR curve to find the optimal decision threshold. Final evaluation used the untouched test set to assess models performance and find the most suitable model for applying SHAP. All experiments were conducted in Python using libraries listed in section 3.8.

4.2 Baseline Model Results

For the Cleveland dataset, nested CV result (Table 4.1) shows that the baseline model LR emerged as one of the three highest-performing models along with SVM and CatBoost, with mean $\pm$ std of the primary metric ROC-AUC = 0.9 ± 0.02 , F1-score = 0.793 ± 0.044 and Sensitivity = 0.783 ± 0.076 ; even outperformed some models that are more advanced like RF. This is because of the preprocessing steps. WOE encoding encoded categorical data into monotonic, log-odds-aligned numeric format, which is literally from LR theory. Numerical preprocessing steps also satisfies LR's core assumption of roughly Gaussian, scaled features.

LR does not need post-hoc probability calibration since it has directly optimized the log-loss, it is built to interpret its output right away as probabilities, so it just got hypertuned and went straight to threshold tuning stage. After having optimal threshold at about 0.4, the

final evaluation still shows LR strength in classification on small dataset with its ability to apply regularization to reduce overfitting, with high ROC-AUC of 0.9361.

Table 4.1. Cleveland Nested CV Performance

Model	Sensitivity	Specificity	ROC-AUC	PR-AUC	F1-score	Precision	Balanced Accuracy
SVM	0.783±0.089	0.870±0.045	0.909±0.014	0.906±0.023	0.807±0.041	0.839±0.036	0.826±0.031
Logistic	0.783±0.076	0.839±0.051	0.900±0.020	0.899±0.025	0.793±0.044	0.808±0.043	0.811±0.035
CatBoost	0.765±0.083	0.885±0.072	0.898±0.027	0.899±0.032	0.804±0.039	0.858±0.069	0.825±0.032
RandomForest	0.756±0.084	0.877±0.069	0.891±0.023	0.895±0.020	0.795±0.023	0.849±0.062	0.817±0.013
XGBoost	0.747±0.106	0.862±0.070	0.887±0.027	0.888±0.034	0.780±0.055	0.830±0.070	0.804±0.041
DecisionTree	0.729±0.074	0.809±0.102	0.819±0.015	0.793±0.039	0.746±0.042	0.777±0.097	0.769±0.039

Table 4.2. Framingham Nested CV Performance

Model	Sensitivity	Specificity	ROC-AUC	PR-AUC	F1-score	Precision	Balanced Accuracy
CatBoost	0.680±0.035	0.656±0.024	0.726±0.035	0.347±0.051	0.378±0.025	0.262±0.020	0.668±0.025
SVM	0.701±0.039	0.629±0.038	0.726±0.037	0.340±0.055	0.372±0.031	0.254±0.025	0.665±0.031
Logistic	0.693±0.035	0.656±0.038	0.730±0.038	0.339±0.056	0.385±0.035	0.266±0.029	0.674±0.033
XGBoost	0.631±0.050	0.683±0.022	0.712±0.035	0.335±0.037	0.371±0.030	0.263±0.023	0.657±0.029
RandomForest	0.256±0.151	0.909±0.091	0.713±0.033	0.332±0.041	0.273±0.076	0.388±0.084	0.582±0.033
DecisionTree	0.602±0.117	0.626±0.066	0.655±0.035	0.237±0.015	0.324±0.031	0.224±0.018	0.614±0.035

As for the Framingham dataset, as expected, the scores are much lower than those of Cleveland. This is due to the noisy nature of the once-longitudinal clinical dataset and its long-term prognosis objective. Although LR is also in the top-3-model list, just like with Cleveland, nested CV (Table 4.2) shows very low results of Precision, PR-AUC and F1-score, compared to what it gained with Cleveland. Final evaluation with the hypertuned LR model shows very slight improvement in Sensitivity, rose from around 0.69 to 0.71 0.72. However, Precision experienced quite a high drop from around 0.34 to only 0.2, making PR-AUC also decrease. All other metrics also decrease to a certain extent. This is the Precision-Recall tradeoff with applying threshold tuning to an imbalanced dataset.

Table 4.3. Final Performance Evaluation of Top 3 Models of Cleveland dataset

Model	Sensitivity	Specificity	ROC-AUC	PR-AUC	F1-score	Precision	Balanced Accuracy
CatBoost	0.9286	0.8182	0.9535	0.9513	0.8667	0.8125	0.8734
SVM	0.9286	0.8485	0.9535	0.9502	0.8814	0.8387	0.8885
Logistic	0.8571	0.9091	0.9361	0.9351	0.8727	0.8889	0.8831

Table 4.4. Final Performance Evaluation of Top 3 Models of Framingham dataset

Model	Sensitivity	Specificity	ROC-AUC	PR-AUC	F1-score	Precision	Balanced Accuracy
CatBoost	0.6899	0.5341	0.6655	0.2670	0.3219	0.2099	0.6120
SVM	0.7287	0.5077	0.6859	0.2747	0.3258	0.2098	0.6182
Logistic	0.7132	0.5661	0.6975	0.3033	0.3452	0.2277	0.6396

4.3 Performance of ML/Ensemble Models

ML and ensemble models demonstrated superior predictive power in the Cleveland dataset, with SVM and CB emerged as top 2 in the best models list after nested CV, achieving a ROC-AUC of 0.909 ± 0.014 and 0.898 ± 0.027 , respectively (Table 4.1). SVM obtained the highest F1-score (0.807 ± 0.041), indicating strong discrimination ability and balanced classification performance. The hypertuned calibrated and threshold-optimized models improved the performance of all best models much higher, jumped from around 0.9-0.91 up to 0.95. All other metrics increased too.

For the Framingham dataset, CB achieved the highest PR-AUC of 0.347 ± 0.051 , while SVM showed comparable performance. Overall, the differences among the top models were relatively small, suggesting that several algorithms were capable of achieving competitive predictive performance.

Overall, the differences among the top models were relatively small, suggesting that several algorithms were capable of achieving competitive predictive performance.

4.4 Calibration Analysis

Calibration quality was assessed using Brier Score: lower values mean better agreement between predicted probabilities and observed outcomes.

Table 4.5. Cleveland Calibration Result - Brier Score

Model	State	Brier Score
SVM	Before Calibration	0.1042
SVM	After Calibration	0.0991
Logistic	Before Calibration	0.0920
Logistic	After Calibration	0.0920
CatBoost	Before Calibration	0.1291
CatBoost	After Calibration	0.1051

For the Framingham dataset, calibration substantially improved CB probability estimates, reducing the Brier Score from 0.204 to 0.125 (Table 4.6). In contrast, SVM showed only minor changes, even higher than before calibration, indicating that its probability estimates were already reasonably calibrated before. From the calibration curves in Figure 4.1, the curves of SVM and CB are very close to the perfect calibration line, especially in the lower range of mean predicted probability.

Table 4.6. Framingham Calibration Result - Brier Score

Model	State	Brier Score
CatBoost	Before Calibration	0.2038

CatBoost	After Calibration	0.1250
SVM	Before Calibration	0.1220
SVM	After Calibration	0.1237
Logistic	Before Calibration	0.2117
Logistic	After Calibration	0.2117

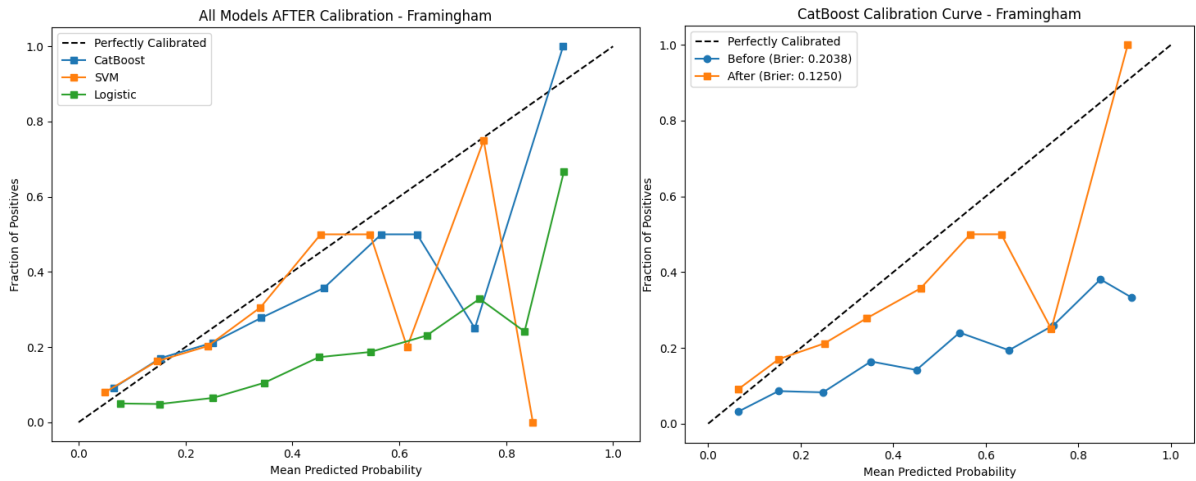


Figure 4.1. Calibration Curves - Framingham

Similarly, for the Cleveland dataset, calibration improved both SVM and CB. The Brier Score of CB decreased from 0.129 to 0.105, while SVM improved from 0.104 to 0.099. The calibration curves (Figure 4.2) also show that, after calibration, the curves move closer to the diagonal lines of perfect calibration than before, especially from the mean predicted probability of about 0.7.

These results indicate that probability calibration was particularly beneficial for CB, improving the reliability of predicted probabilities used in subsequent threshold optimization and clinical decision-making.

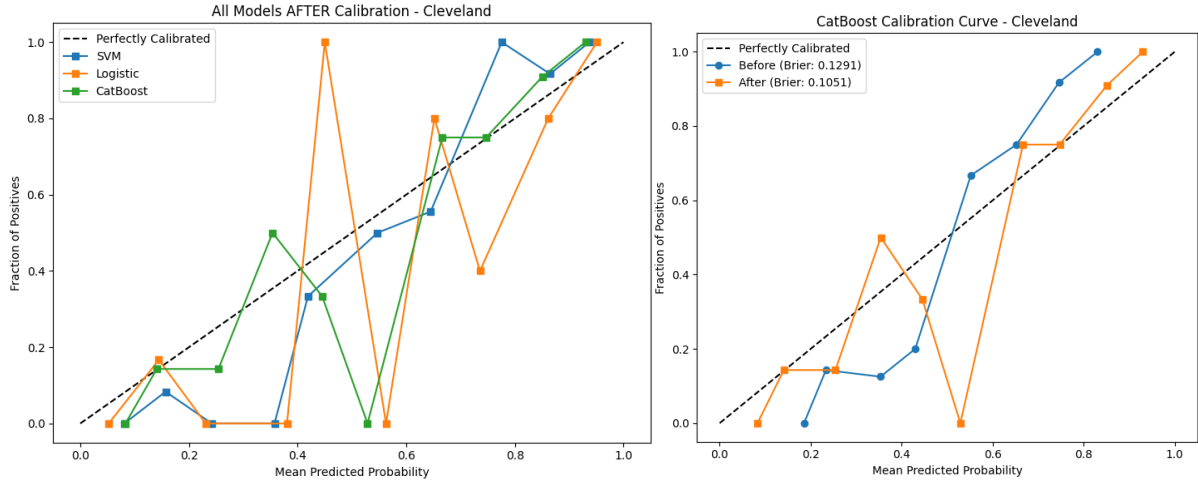


Figure 4.2. Calibration Curves - Cleveland

4.5 Threshold Optimization

The optimized thresholds differed substantially from the conventional value of 0.5. For example, CB and SVM obtained thresholds of approximately 0.11 on the Framingham dataset. This indicates that calibrated probabilities were concentrated at lower values, making the default threshold unsuitable for clinical risk prediction, while for Cleveland dataset, the optimal thresholds of SVM and CB are both higher than 0.5. This is shown very clearly in the calibration curves in the previous section.

After all the analyses, CB was chosen to be the model that would proceed to the explanations stage. One other reason contributed to this selection is because the TreeSHAP that supports CB is generally much more stable than KernelSHAP used for SVM. Since the study evaluates explanation consistency and stability through bootstrap analysis, a deterministic explanation framework was preferred to reduce variability introduced by the explanation method itself.

4.6 Explainability Results

4.6.1 Baseline Model Interpretation

Before interpreting the CatBoost models using SHAP, the LR model was first examined because its coefficients can be interpreted directly.

Table 4.7. Logistic Regression Models' Structures of Cleveland (left) and Framingham (right)

Feature	Coefficient	Feature	Coefficient
ca	1.7475	age	1.0486
sex	0.8858	male	0.9338
cp	0.6617	cigsPerDay	0.8897
slope	0.6255	diabetes	0.6531
thal	0.5736	prevalentStroke	0.6338
restecg	0.3805	BPMeds	0.4654
trestbps	0.3566	education	0.3897
chol	0.3446	sysBP	0.3183
exang	0.3028	prevalentHyp	0.2143
oldpeak	-0.0285	totChol	0.1252
age	-0.1392	diaBP	0.0623
fbs	-0.3146	glucose	0.0358
thalach	-0.4825	BMI	0.0183
		heartRate	-0.0410
		currentSmoker	-4.5799

For the Cleveland dataset (Table 4.7 left), the largest positive coefficients were with features: number of major vessels (ca), sex, chest pain type (cp), ST segment slope (slope), and exercise thallium scintigraphic defect (thal). This suggests that these variables contributed the most to increasing the probability of heart disease. On the other hand, maximum heart rate (thalach) had a negative coefficient, indicating that patients with higher maximum heart rates were generally predicted to have a lower risk of CHD.

For the Framingham dataset, age had the largest positive coefficient, followed by gender (male), smoking status (cigsPerDay) and history of diabetes/stroke (diabetes and prevalentStroke). These results agree with well-known CVD risk factors mentioned in Chapter 2 before.

4.6.2 Global Feature Contribution

As mentioned, since LR assumes a linear relationship between each feature and the prediction, it may not fully capture the complex interactions among clinical variables.

To transition the model from a “black box” to a clinically interpretable tool, SHAP values were calculated for the CB model.

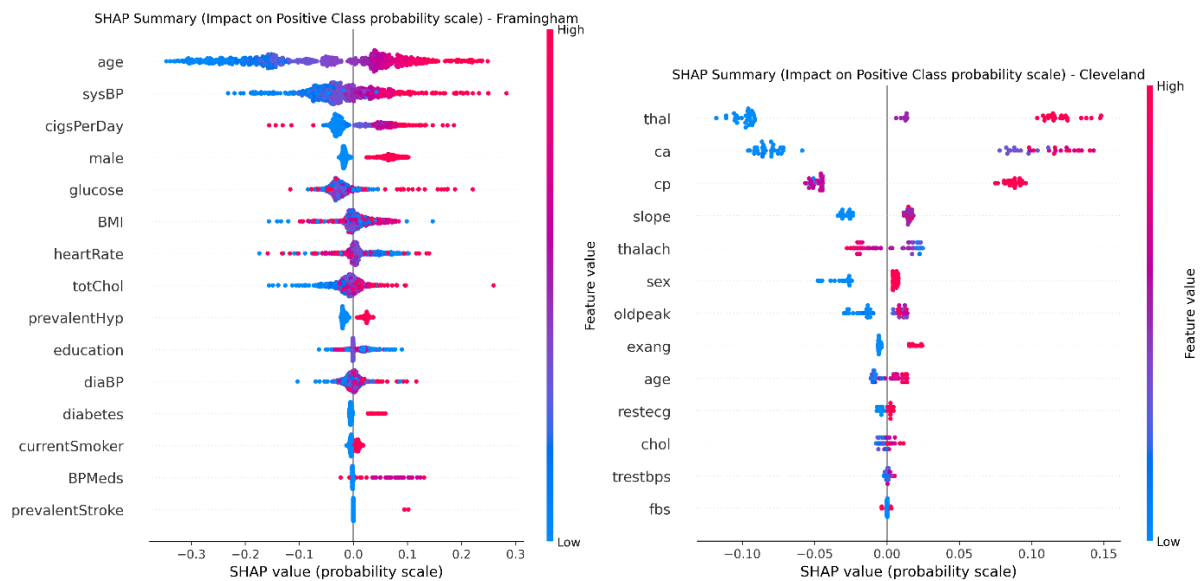


Figure 4.3. Framingham (left) and Cleveland (right) SHAP Beeswarm plots for Global Explainability

The SHAP beeswarm plot (Figure 4.3) serves as the primary visual evidence for this global interpretability. In this visualization, clinical features are ranked vertically based on their total influence on the model’s decisions, with the most dominant predictors being at the top. The horizontal distribution of points illustrates how specific values for these features shift the probability of a CVD prediction. Points extending to the right indicate an increase in predicted risk, while those to the left signify a protective effect or a lower likelihood of disease. The accompanying color gradient further clarifies these relationships, with red representing high feature values and blue representing low values. For instance, the clustering of red points on the positive side for a feature like “age” (Figure 4.3 left) confirms that the model has successfully learned to associate older people with an increased probability of illness.

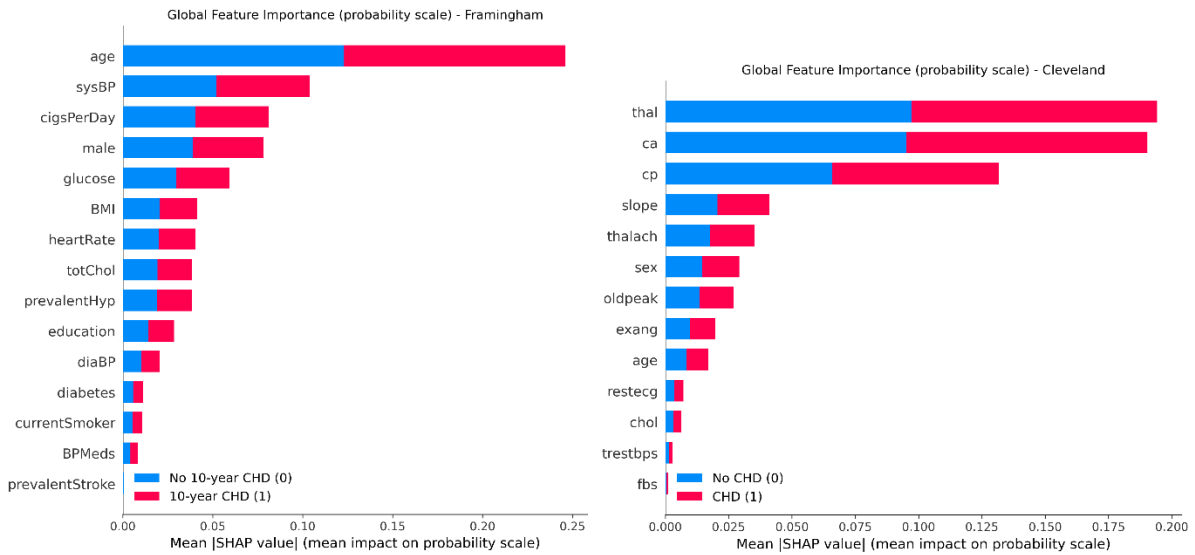


Figure 4.4. Framingham (left) and Cleveland (right) SHAP Bar plots for Global Explainability

Beside Beeswarm plot, another visualization that also provides explainability for the model’s prediction is bar plot. The bar plots in Figure 4.4 show the same order of feature influence as the beeswarm plot above.

It is noteworthy that the four most influence features – “age”, “sysBP”, “cigsPerDay” and “male” of Framingham and “thal”, “ca”, “cp” of Cleveland – show very obvious difference feature characteristics between two scenarios. The most important features of prognosis task are demographic (e.g., age and gender) and lifestyle factors (e.g., cigsPerDay and sysBP). For the diagnosis task, the top features are clinical results (e.g., chest pain type “cp”) and physiological/noninvasive tests (e.g., “thal” and “ca” are most influential ones).

For prognosis, the model focuses on “who the patient is” rather than “how they feel right now”. This explains the lower performance mentioned earlier, lifestyle factors are markers of risk, but they do not guarantee a future event. Opposed to Framingham, “age” is much lower on the list of Cleveland. This means that in diagnosis, the model ignores general lifestyle and looks for physical proof that the heart is currently struggling.

4.6.3 Local Patient-Level Risk Explanations

Local Waterfall and Force plots (Figure 4.5) provided a transparent view of individual patient risk, showing how factors work together to push a prediction toward a positive prediction.

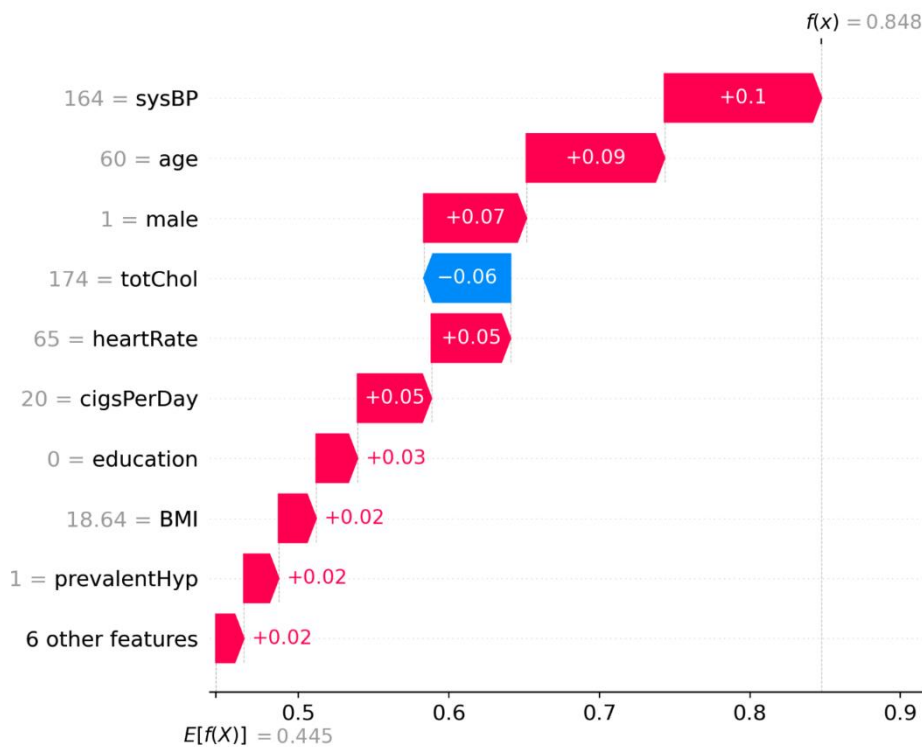


Figure 4.5. Force (above) and Waterfall (below) Plots for patient-level interpretation of Framingham

As observed from the plots of Framingham, the patient is predicted to have CHD with a probability of 84.8%, which is above the baseline of 44.5%. This is because many other factors have more impact to the prediction, especially high age male gender and high systolic BP – the factors that show highest influence from the global explanation plots.

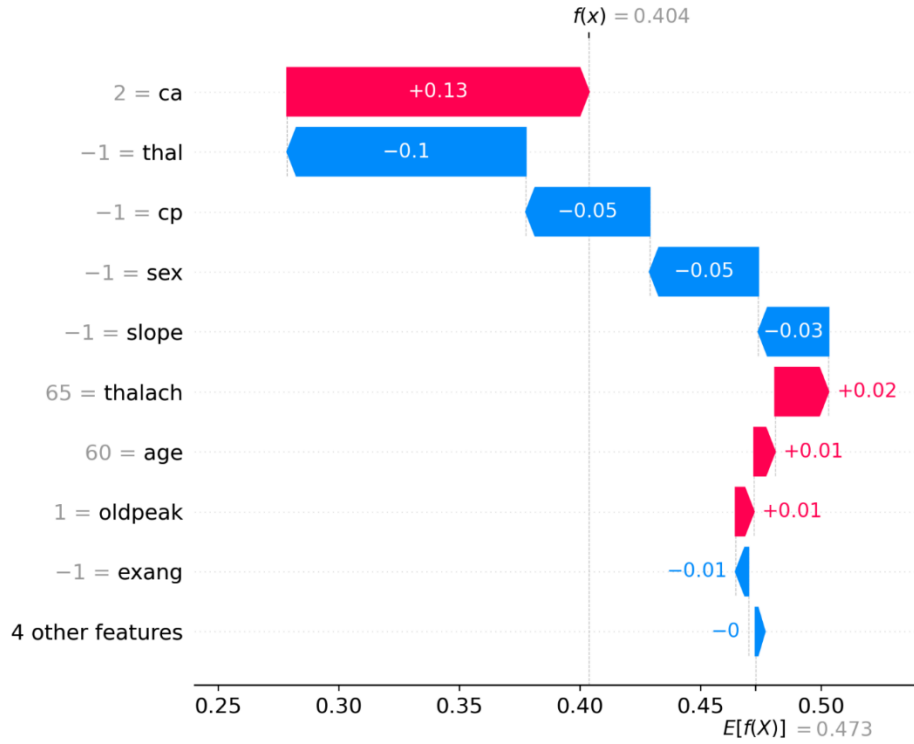


Figure 4.6. Force (above) and Waterfall (below) Plots for patient-level interpretation of Cleveland

Having the same age, gender, heart rate, cholesterol level as the patient above, however, the patient here in diagnosis task received a negative result. In diagnosis task of Cleveland, age, gender are not import as testing result like “ca”, “thal”.

4.7 SHAP Consistency and Stability

To compensate for the lack of external validation, this research employs Bootstrap Stability Analysis and Accumulated Local Effects (ALE) to quantitatively verify the internal consistency and robustness of the explainability module – SHAP.

As mention in section 3.6.2, for each numeric feature, the blue scatter (SHAP, left axis) and red line (ALE, right axis) should follow the same shape and direction. As illustrated in Figure 4.7, for the Framingham dataset, “age” and “sysBP” show a clear positive slope in both methods, which is strong consistency, as those two are two most important numerical features from the SHAP global plots (Figure 4.7).

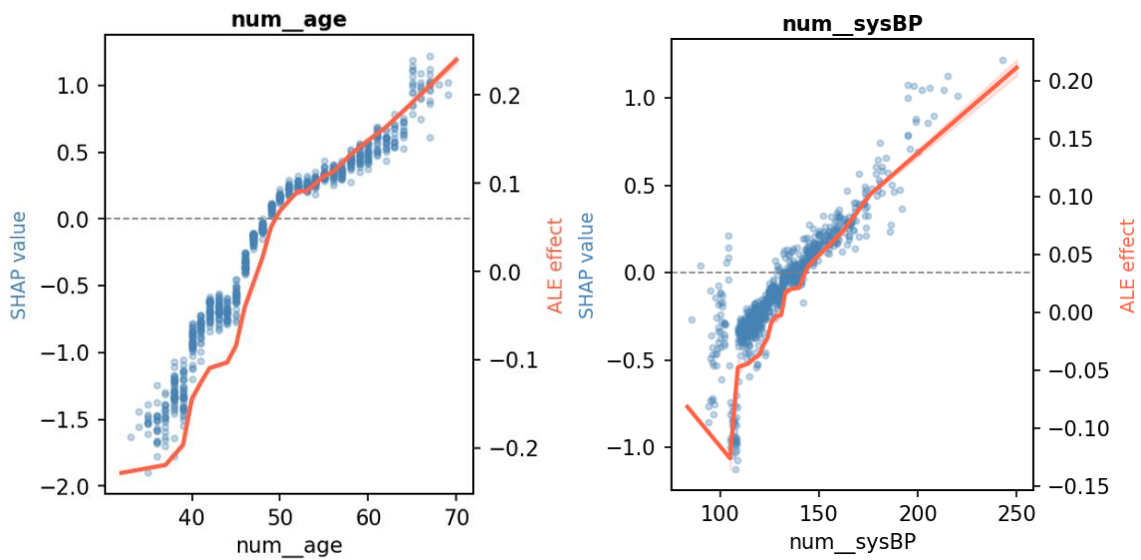


Figure 4.7. Framingham SHAP Dependence vs. ALE - Global Feature Effects

The result of Framingham Bootstrapping shows extreme stability of the results (Figure 4.8). “age” has rank CV = 0%, which means it was rank 1 in every single one of the 20 bootstrap runs, as well as the four most important features ranking exactly the same as the SHAP global plots.

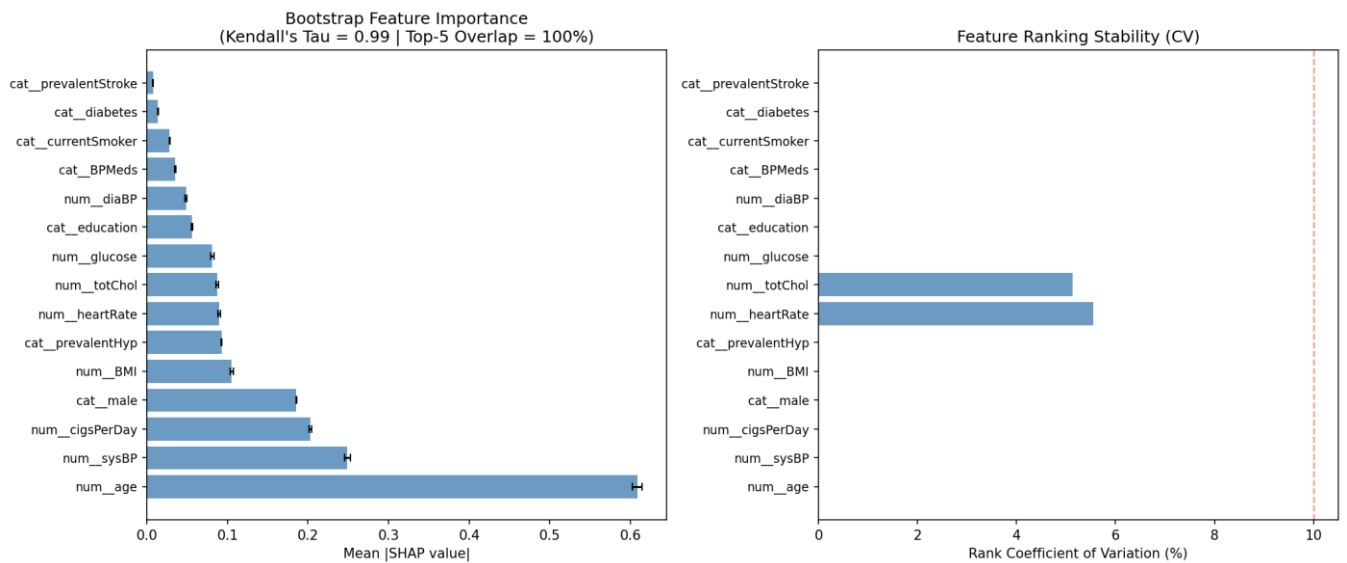


Figure 4.8. Framingham SHAP Bootstrap Stability

As for the Cleveland dataset, two top numerical feature “oldpeak” and “thalach” have strongest consistency, both show the clearest agreement between SHAP and ALE (Figure 4.9). This ALE consistency checking method just considered numerical features for clearer patterns, so even though “oldpeak” is just moderately important, the plot still show that the explanations are consistent for this particular feature.

In the Bootstrap plots (Figure 4.10), opposed to Framingham, the least important features of Cleveland are perfectly stable with near-zero CV, top feature “cp” and “thal” are also very stable. What concerning is the positions of “sex” (rank 3) “thalach” (rank 4), and “slope” (rank 5). Although their positions are perfectly correct comparing to the SHAP global plots, but their CV are high. This is because CV is calculated as standard deviation/mean, and when a feature’s mean rank is 3 or 4, even a tiny standard deviation of 0.2 (meaning it occasionally swaps to the other rank) produces a large CV percentage. Therefore, they are not exactly unstable, they are consistently at the top of the list, just sometimes switching positions with one another. Cleveland has this situation with stability is reasonable, since its size is too small, after the split, the test set has only 61 samples, which means bootstrap subsample is very small, so the ranking is unstable is normal.

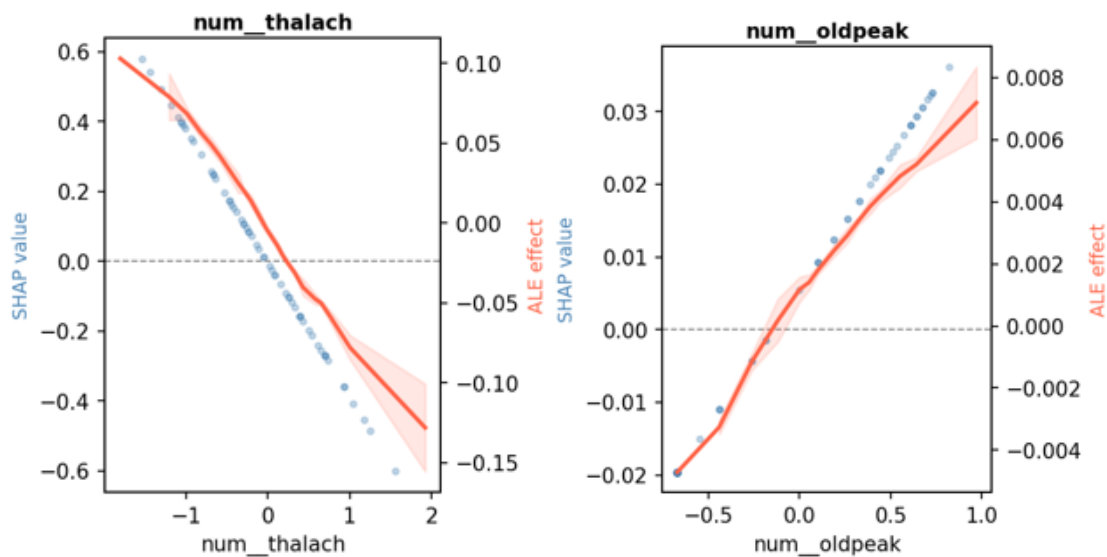


Figure 4.9. Cleveland SHAP Dependence vs. ALE - Global Feature Effects

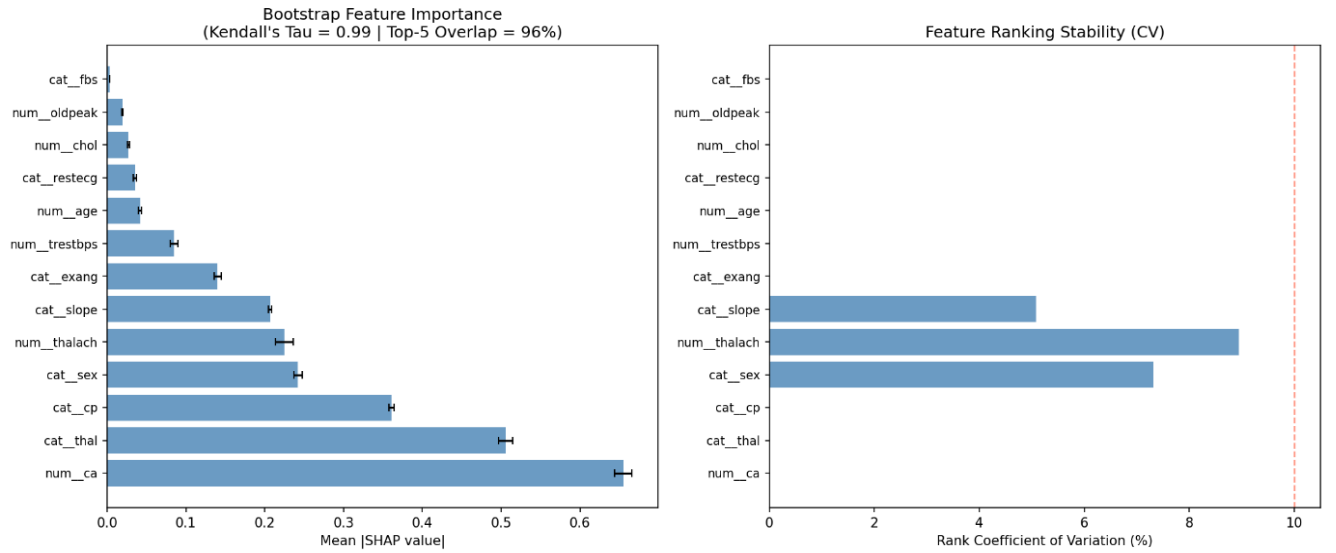


Figure 4.10. Cleveland SHAP Bootstrap Stability

In addition, Kendall's Tau (τ) was used to quantify the agreement between feature rankings obtained across repeated bootstrap runs. Kendall's Tau is a rank correlation coefficient ranging from -1 to 1:

$\tau = 1$ indicates perfect agreement between rankings.

$\tau = 0$ indicates no association.

$\tau = -1$ indicates complete disagreement.

Higher values indicate stronger agreement in feature rankings across bootstrap samples. The mean Kendall's Tau obtained from 20 bootstrap iterations was used as a global measure of explanation consistency. The high mean Kendall's Tau of 0.99 in both datasets' Bootstrap SHAP means that the relative importance of features remained the same across repeated analyses.

One other method for validating stability is Top-K Overlap. It measures how frequently the most important features remain within the top-ranked positions across bootstrap runs. The perfect 100% Top-5 Overlap percentage indicates that for the Framingham dataset, the same features consistently appear among the most influential predictors. Cleveland also achieved very high percentage of 96%.

4.8 Discussion of Findings

A primary finding of this research is the significant performance gap between the Cleveland and Framingham datasets. The high scores achieved in Cleveland result tables suggest that diagnosing current CVD is highly achievable with ML, while those in the Framingham dataset reflects the inherent difficulty of long-term prognosis. The “noise” in human behavior over a decade makes it harder for any model to achieve the same precision as a diagnostic model.

The results also highlight the importance of probability calibration and threshold optimization. Before threshold tuning, CB and SVM produced extremely low numbers of positive predictions on the Framingham test set, resulting in poor Sensitivity and F1-scores despite maintaining reasonable ROC-AUC and PR-AUC values. This indicates that the default classification threshold of 0.5 was inappropriate after calibration.

The threshold tuning step highlights the trade-off between Sensitivity and Precision in healthcare. Prioritizing a high Recall ensures the model catches as many at-risk patients as possible, even if the Precision got lower. This is vital in medical prevention where missing a sick patient is much more dangerous than correctly identify healthy person.

SHAP proves that the models follow logical medical reasoning. The Framingham model focuses on long-term lifestyle risks like age and smoking, while the Cleveland model prioritizes immediate clinical tests like chest pain type and test results.

Compared to Umar et al. (2025) study, of which one of the datasets used is the same as this study (they marked the Cleveland dataset as D3), the following findings were obtained: Both studies obtained high result with SVM and RF models. Though this study’s main metrics of comparison is ROC-AUC, differed from Umar et al. – Accuracy, the overall metrics results of this study for SVM model still higher than theirs. This difference in value is probably because of difference in data preprocessing and cleaning methods (since this study has tried some methods that are not commonly used), as well as different hyperparameters used for tuning and tuning method (this study used Randomized Search, after a whole process of nested CV for model selection, while Umar et al. used Grid Search).

As for the Framingham dataset, most papers reviewed have much higher performance than this study’s. The Framingham was expected to not have high performance in its

prediction models due to its noisy nature. However, the reason for this gap in performance between this study and the other papers must be the lack of feature selection and dimensional reduction methods, which these papers have. One small finding, especially in the two papers Suhatri et al. (2024) and Sadhukhan et al. (2025) whose models' performance are so high, is that they did not have the "education" column in their Framingham dataset. This could be one reason why they could achieve results that high, along with the differences in preprocessing and handling imbalance data. In terms of explainability, this paper and Sadhukhan et al. (2025) came to the result that for this particular dataset, age, gender and blood pressure are top features contribution to the explanations of predictions.

Table 4.8. Umar et al. (2025) Models' Performance of Cleveland dataset

Model	F1-score	Recall	Accuracy	Precision	ROC-AUC
SVM	0.8628	0.8980	0.8462	0.8302	0.8985
RF	0.8387	0.7959	0.8352	0.8864	0.9218

To validate the reliability of these explanations, two additional analyses were conducted. Bootstrap stability analysis showed that SHAP feature rankings remained largely consistent across 20 random subsamples of the test data, with most features maintaining a stable rank and low coefficient of variation. Cross-method consistency was further examined by comparing SHAP with Accumulated Local Effects (ALE) plots. Together, these results suggest that the SHAP explanations produced in this study are not only meaningful in a clinical context, but are also stable and consistent with an independent XAI method. These findings strengthen confidence in the reliability of the generated explanations and address a limitation commonly observed in previous XAI studies, where explanations are presented without validation.

Overall, the findings show that accurate and interpretable CVD prediction can be achieved by combining ML models with explainable AI techniques. More importantly, the results demonstrate that not only predictive performance is evaluated, but explanation reliability should also be evaluated alongside, particularly in healthcare applications where transparency and trustworthiness are critical.

Chapter 5 . CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

This study investigated the application of Explainable Artificial Intelligence (XAI) in cardiovascular disease prediction across two different clinical scenarios: diagnosis using the Cleveland Heart Disease dataset and prognosis using the Framingham dataset.

The experimental results demonstrated that machine learning (ML) models can effectively predict cardiovascular outcomes in both settings. Models achieved stronger predictive performance on the Cleveland dataset than on the Framingham dataset, reflecting the greater difficulty of long-term risk prediction compared to disease diagnosis. Probability calibration and threshold optimization further improved the clinical usefulness of model predictions, particularly for the imbalanced Framingham dataset.

The explainability analysis identified the most influential risk factors for each prediction task and showed noticeable differences between diagnostic and prognostic prediction. Furthermore, the consistency and stability analyses showed that the generated explanations were generally reliable, with SHAP explanations demonstrating agreement with ALE and remaining relatively stable across bootstrap samples.

Overall, the findings suggest that combining ML with explainable AI (XAI) can improve both predictive performance and interpretability in cardiovascular disease prediction.

5.2 Contributions of the Research

The first contribution of this thesis is providing a comparative analysis between diagnostic and prognostic cardiovascular prediction tasks. The results demonstrate that different prediction objectives may lead to different influential risk factors and model behaviors.

Secondly, the study proposes a ML framework that integrates nested cross-validation, calibration, threshold optimization, and explainability analysis for cardiovascular disease prediction.

Thirdly, SHAP was applied to provide both global and local explanations of model predictions, improving the transparency of ML models.

Finally, the study extends beyond conventional XAI applications by evaluating explanation reliability through ALE comparison, bootstrap SHAP analysis, Kendall's Tau, and Top-K overlap metrics. This provides additional evidence regarding the consistency and stability of the generated explanations.

5.3 Limitations of the Study

Despite its contributions, this study has several limitations.

First, the study relies on two publicly available benchmark datasets, particularly the Cleveland dataset, which contains a relatively small number of samples.

Second, no external validation dataset was available, limiting the generalizability of the findings to other clinical populations.

Finally, although SHAP and ALE were used to improve model interpretability, these methods explain predictive relationships, not causal relationships. Therefore, the generated explanations should be interpreted as decision-support information rather than clinical evidence of causation.

5.4 Future Research Directions

First, future research should evaluate the proposed framework using larger and more diverse clinical datasets collected from different healthcare institutions or official clinical sources. External validation would provide stronger evidence regarding the generalizability and robustness of both the predictive models and the generated explanations.

Second, additional data types such as electrocardiograms, medical imaging, laboratory time-series data, and clinical notes could be incorporated. Combining structured and

unstructured medical data may improve predictive performance and provide a more comprehensive understanding of cardiovascular disease risk.

Third, future studies may investigate more advanced ML and DL architectures. Comparing explanation reliability across traditional ML models and DL models could provide further insight into the trade-off between predictive performance and interpretability.

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